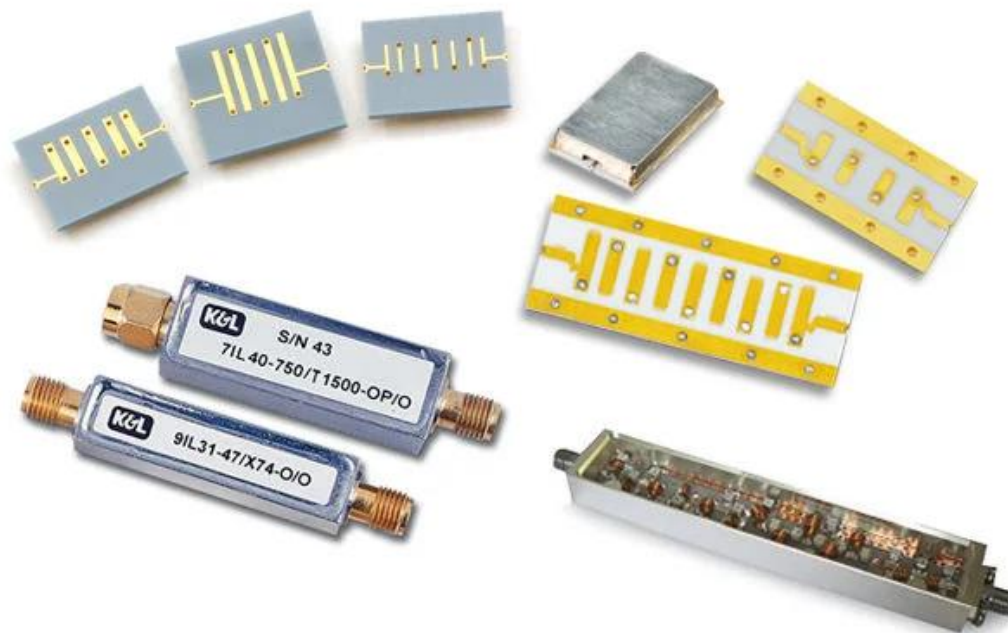


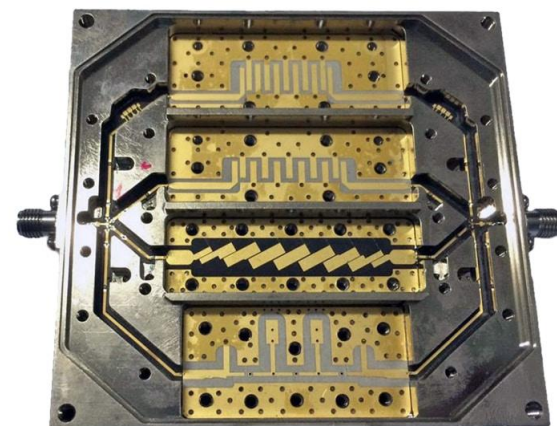
第二章 滤波器设计

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2026-04



滤波器

- 滤波器是用来控制频率响应的二端口器件
- 滤波器分为低通、高通、带通、带阻滤波器
- 滤波器理论开始于二战之前
- 镜像参数法滤波器设计用于低频滤波器设计
- 微波滤波器主要基于插入损耗法进行设计
- 目前滤波器设计软件较为成熟 (CAD)
- 微波滤波器可以由L/C、传输线等构成



滤波器的特性

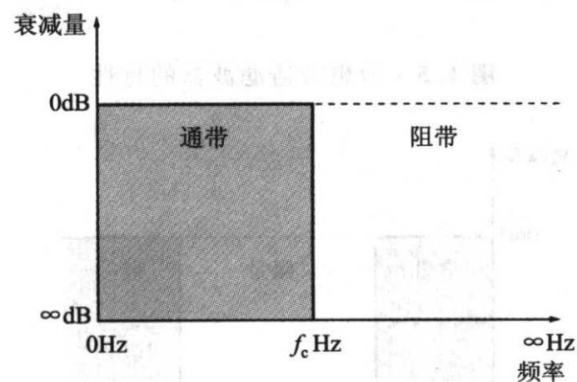


图 1.3 理想低通滤波器的特性

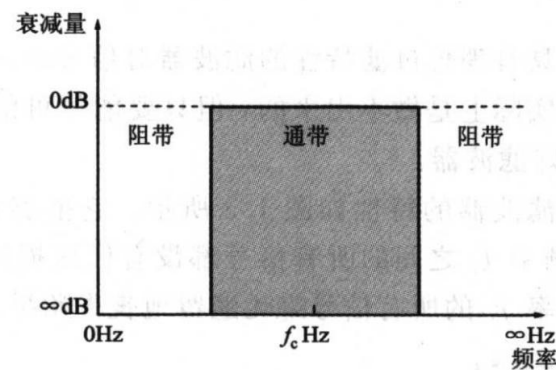


图 1.5 理想带通滤波器的特性

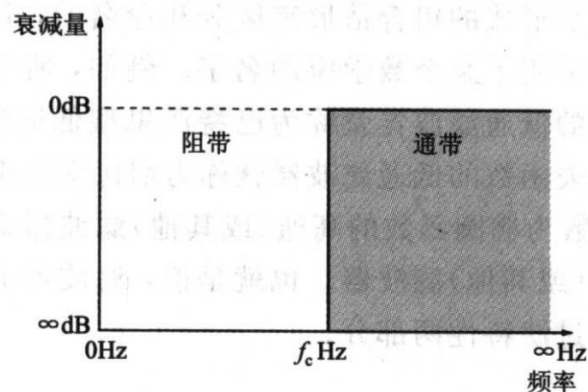


图 1.4 理想高通滤波器的特性

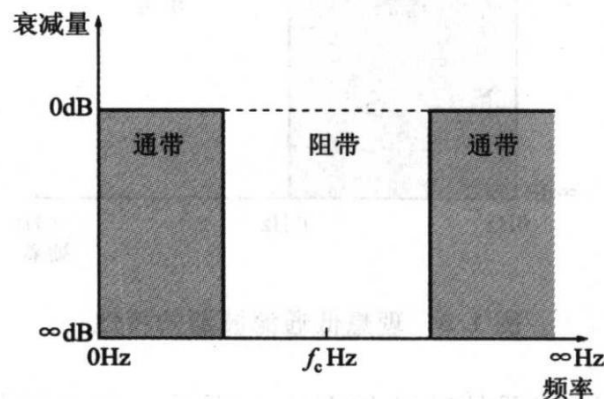
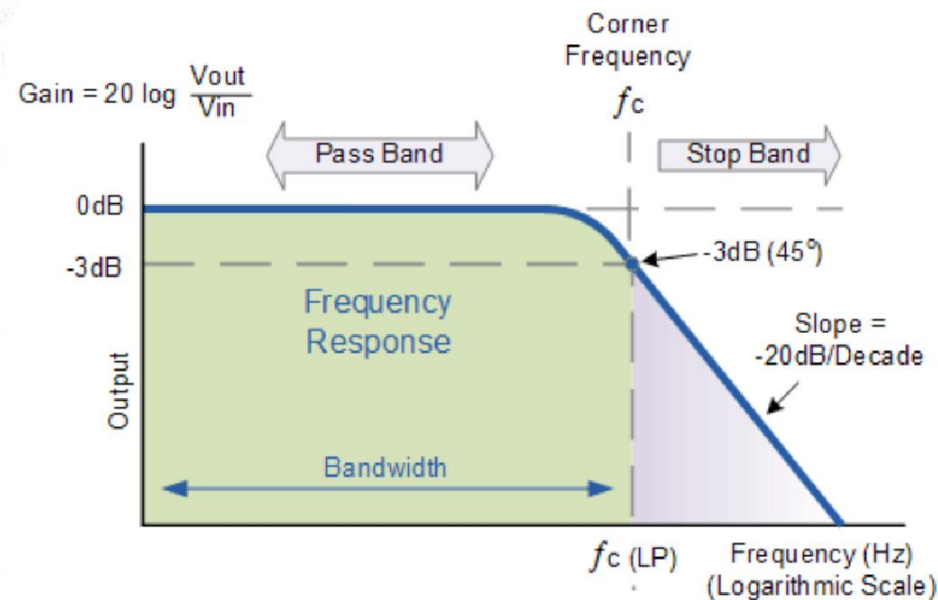
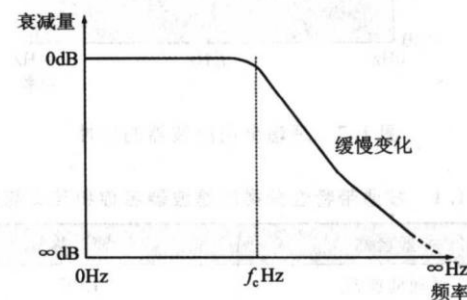
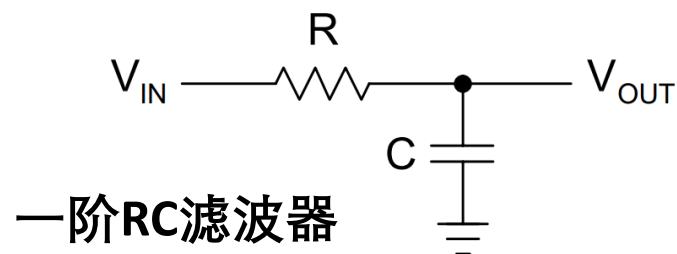


图 1.6 理想带阻滤波器的特性

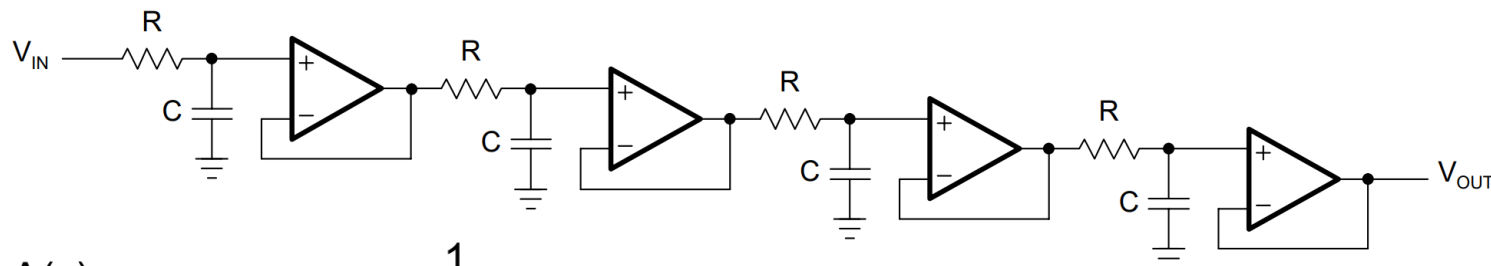


回顾：低频滤波器



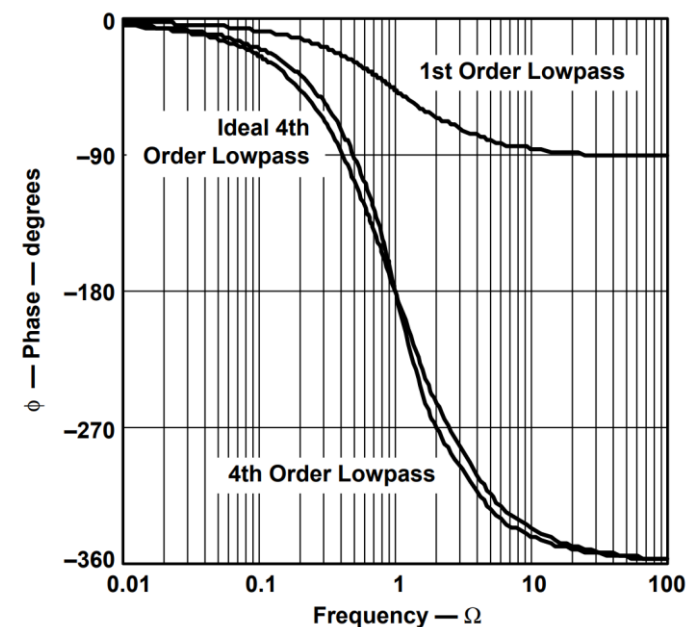
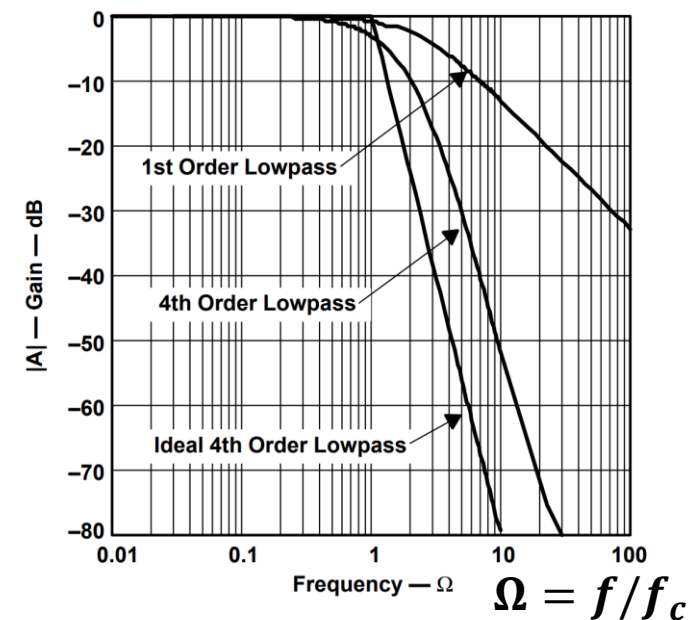
$$A(s) = \frac{\frac{1}{RC}}{s + \frac{1}{RC}} = \frac{1}{1 + sRC}$$

$$s = j\omega + \sigma$$



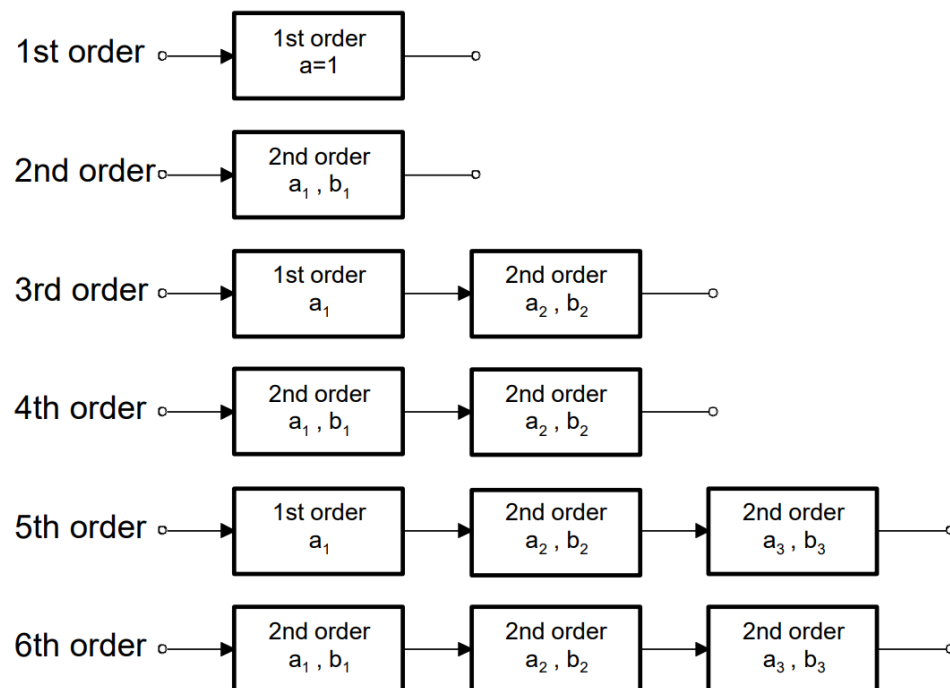
$$A(s) = \frac{1}{(1 + \alpha_1 s)(1 + \alpha_2 s) \dots (1 + \alpha_n s)}$$

$$A(s) = \frac{A_0}{(1 + a_1 s + b_1 s^2)(1 + a_2 s + b_2 s^2) \dots (1 + a_n s + b_n s^2)} = \frac{A_0}{\prod_i (1 + a_i s + b_i s^2)}$$

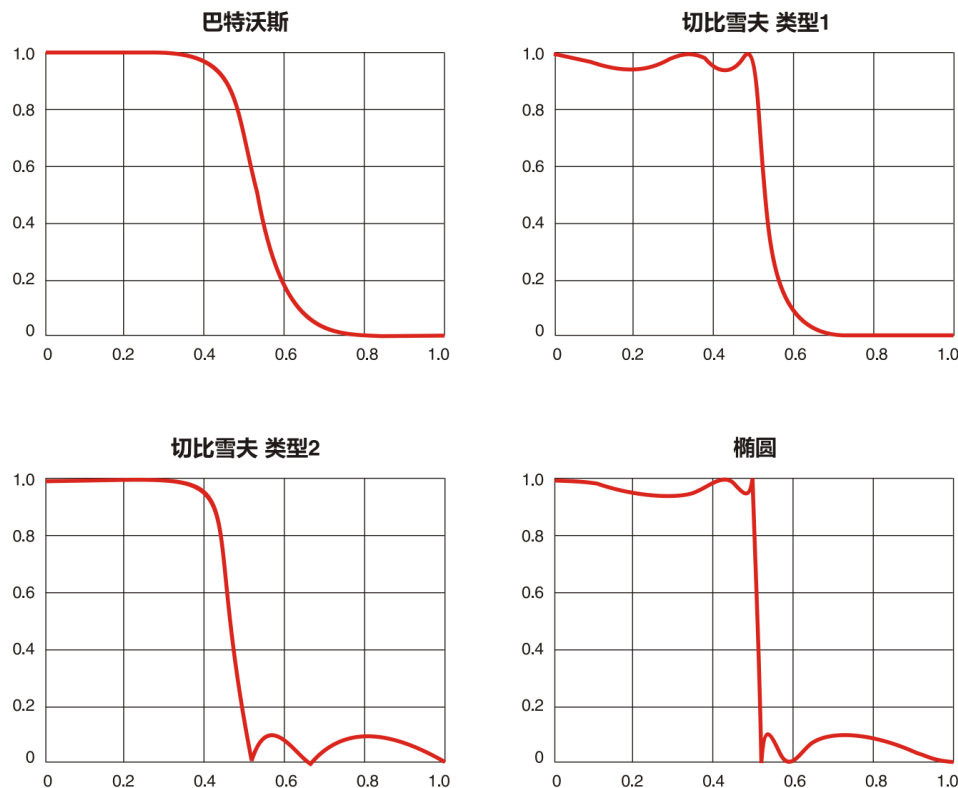


回顾：低频滤波器

$$A(s) = \frac{A_0}{(1 + a_1s + b_1s^2)(1 + a_2s + b_2s^2) \dots (1 + a_ns + b_ns^2)} = \frac{A_0}{\prod_i (1 + a_is + b_is^2)}$$



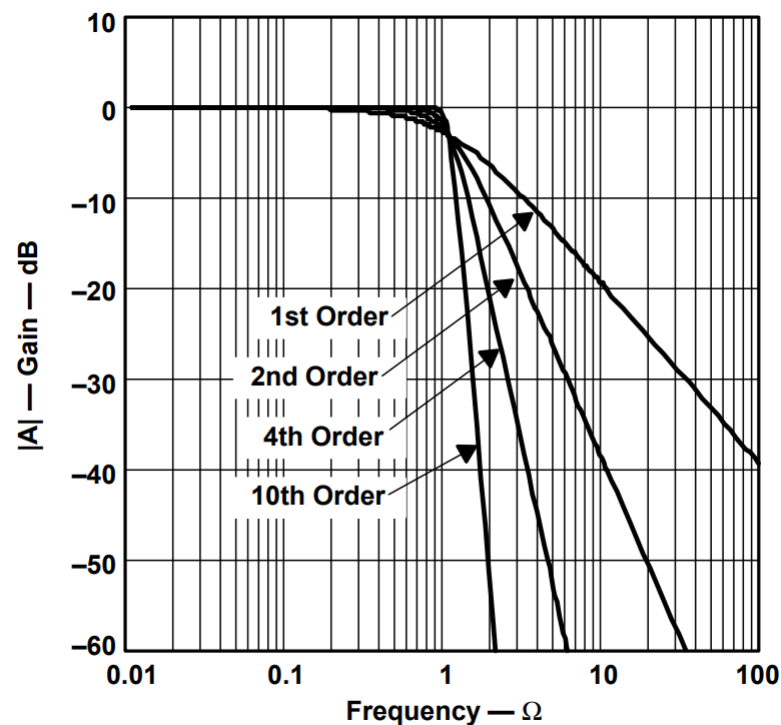
滤波器的特性



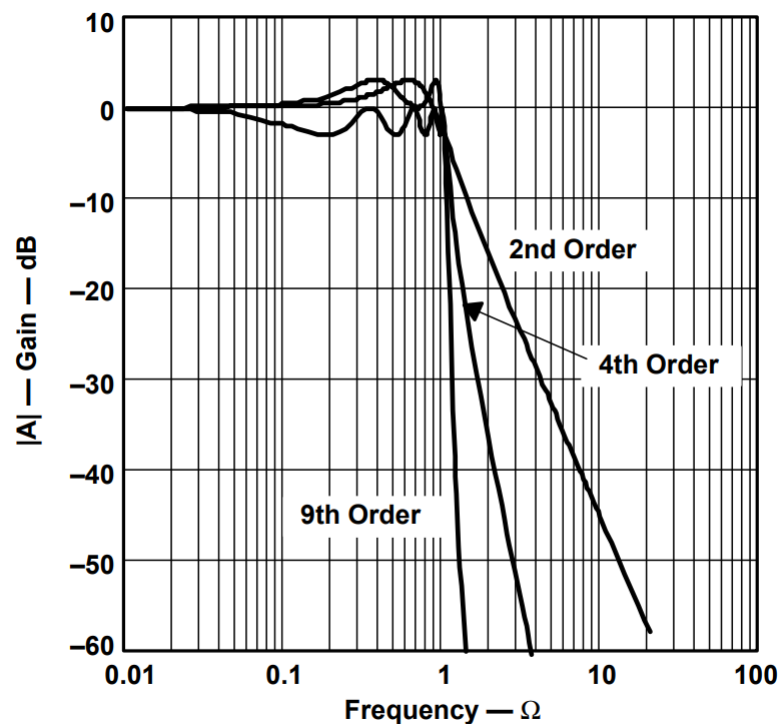
- 无源RC滤波器的传递函数无法实现进一步优化，因其缺乏复数极点。
- 若要利用无源元件产生共轭复数极点，唯一的方法是采用LRC滤波器。这类滤波器主要应用于高频场景。
- 在较低频率范围（<10 MHz）内，所需的电感值会非常大，导致滤波器制造成本过高而不经济。因此在这些情况下，通常采用有源滤波器。

巴特沃斯型 瓦格纳(Wagner)	通带内响应最为平坦，偶尔也称为power term filter
切比雪夫型 (等波纹滤波器)	截止特性特别好；群延时特性不太好；通带内有等波纹起伏
逆切比雪夫型 (巴特沃斯-切比雪夫滤波器)	阻带内有零点(陷波点)。由于椭圆型比它能得到更好的截止特性，因而它不太使用
椭圆函数型 (联立切比雪夫滤波器)	通带内有起伏，阻带内有零点。截止特性比其他滤波器都好，但对器件要求严
贝塞尔型 (延时最平坦滤波器)	通带内延时特性最平坦，截止特性相当差
高斯型	这种函数型的BPF常用于决定频谱分析仪带宽的滤波器中
相位等波纹型	通带内的相位是等波纹变化的
勒让德型	截止特性比巴特沃斯型好，并且可以用小的器件值来实现
近代设计法	
古典设计法	
定K型	设计简单差，易于增加阶数
m推演型	能得到比定K型更陡峭的截止特性，但阻带特性差

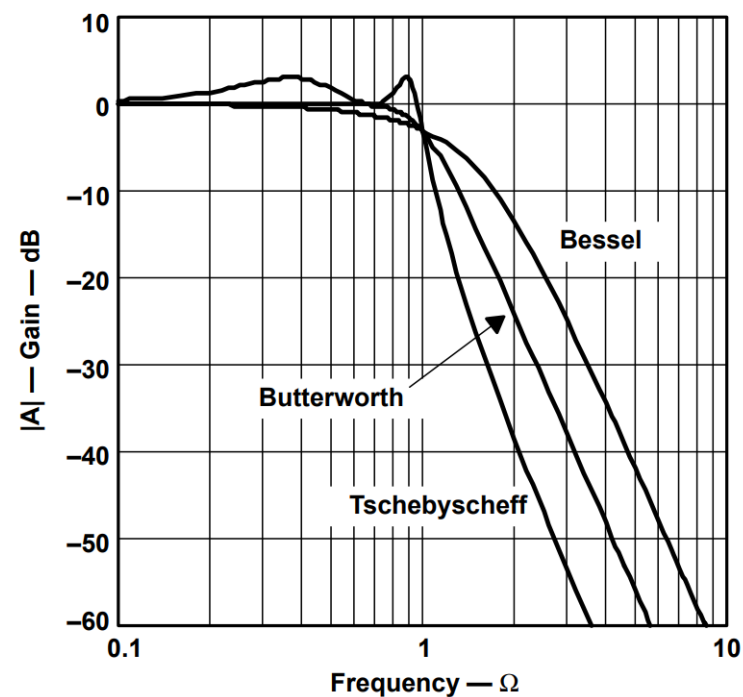
不同滤波器的特点



巴特沃斯滤波器

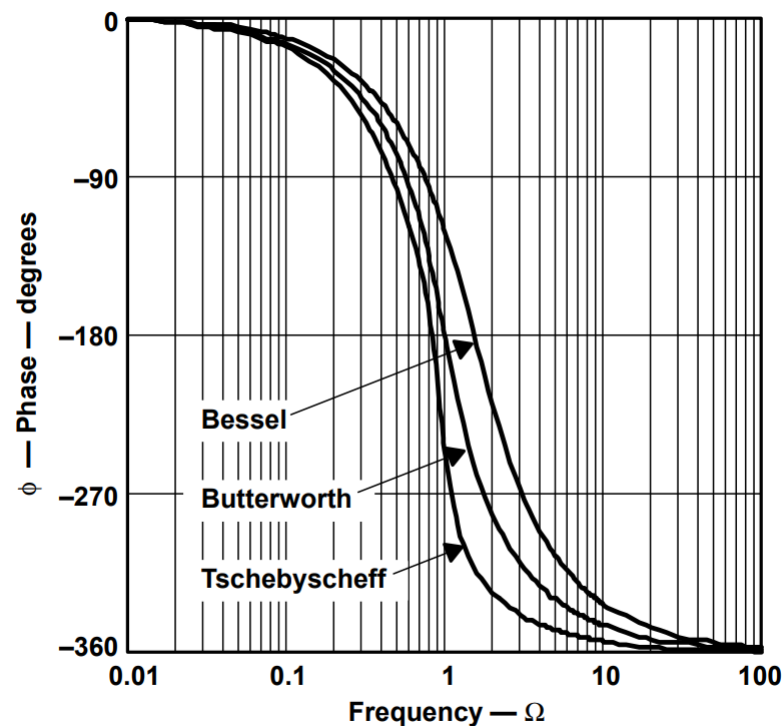


切比雪夫滤波器

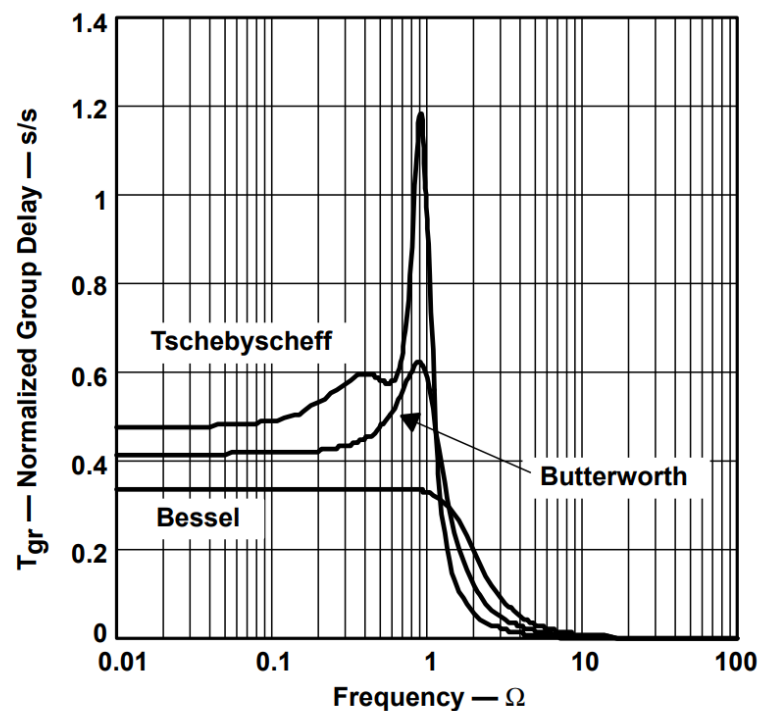


不同滤波器特点

不同滤波器的特点

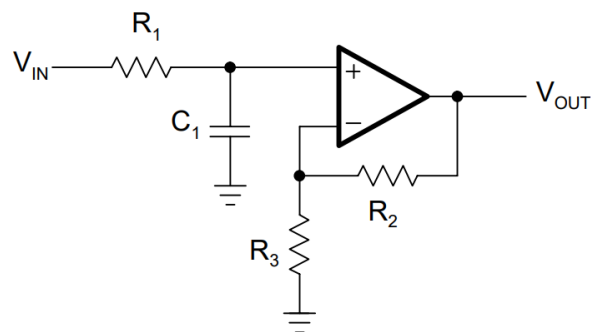


相位比较



群延时比较

低频有源滤波器



同相一阶低通滤波器

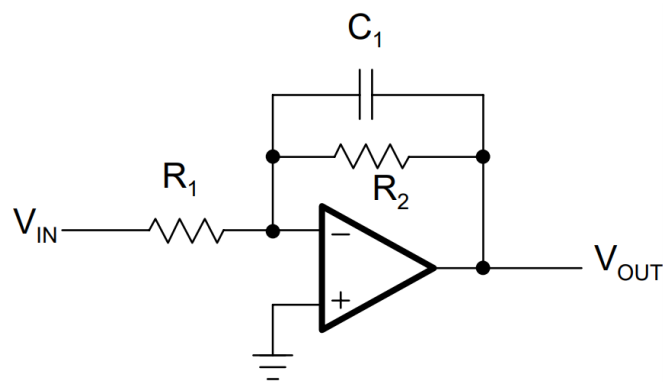
$$V_- = \frac{V_{out} R_3}{R_2 + R_3}$$

$$V_+ = V_- = \frac{V_{in} \cdot 1/sC_1}{R_1 + 1/sC_1}$$

$$A(s) = \frac{1 + R_2/R_3}{1 + j\Omega\omega_c R_1 C_1}$$

$$A_0 = 1 + \frac{R_2}{R_3}$$

$$a_1 = \omega_c R_1 C_1$$



反相一阶低通滤波器

$$A(s) = \frac{-R_2/R_1}{1 + j\Omega\omega_c R_2 C_1}$$

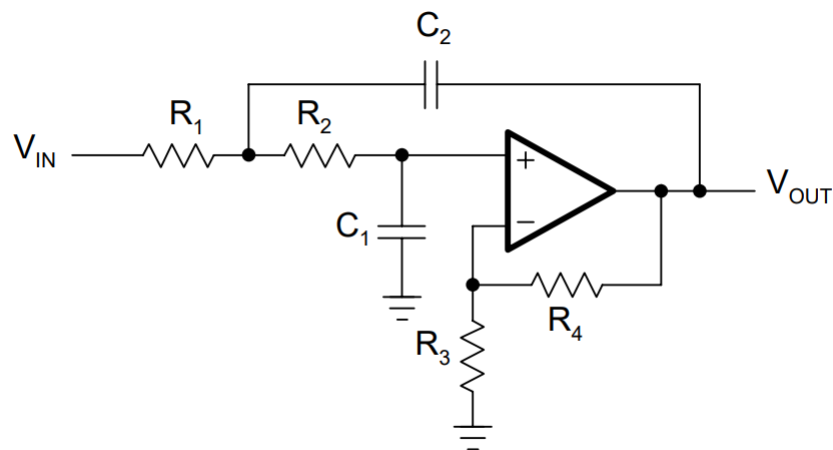
$$A_0 = -\frac{R_2}{R_1}$$

$$a_1 = \omega_c R_2 C_1$$

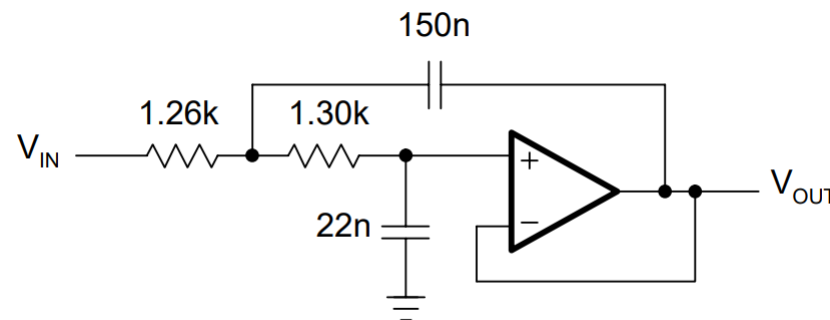
Table 16-4. Bessel Coefficients

n	i	a _i	b _i	k _i = f _{ci} /f _c	Q _i
1	1	1.0000	0.0000	1.000	—
2	1	1.3617	0.6180	1.000	0.58
3	1	0.7560	0.0000	1.323	—
	2	0.9996	0.4772	1.414	0.69
4	1	1.3397	0.4889	0.978	0.52
	2	0.7743	0.3890	1.797	0.81
5	1	0.6656	0.0000	1.502	—
	2	1.1402	0.4128	1.184	0.56
	3	0.6216	0.3245	2.138	0.92
6	1	1.2217	0.3887	1.063	0.51
	2	0.9686	0.3505	1.431	0.61
	3	0.5131	0.2756	2.447	1.02
7	1	0.5937	0.0000	1.648	—
	2	1.0944	0.3395	1.207	0.53
	3	0.8304	0.3011	1.695	0.66
	4	0.4332	0.2381	2.731	1.13
8	1	1.1112	0.3162	1.164	0.51
	2	0.9754	0.2979	1.381	0.56
	3	0.7202	0.2621	1.963	0.71
	4	0.3728	0.2087	2.992	1.23
9	1	0.5386	0.0000	1.857	—
	2	1.0244	0.2834	1.277	0.52
	3	0.8710	0.2636	1.574	0.59
	4	0.6320	0.2311	2.226	0.76
	5	0.3257	0.1854	3.237	1.32
10	1	1.0215	0.2650	1.264	0.50
	2	0.9393	0.2549	1.412	0.54
	3	0.7815	0.2351	1.780	0.62
	4	0.5604	0.2059	2.479	0.81
	5	0.2883	0.1665	3.466	1.42

低频有源滤波器



二阶低通滤波器

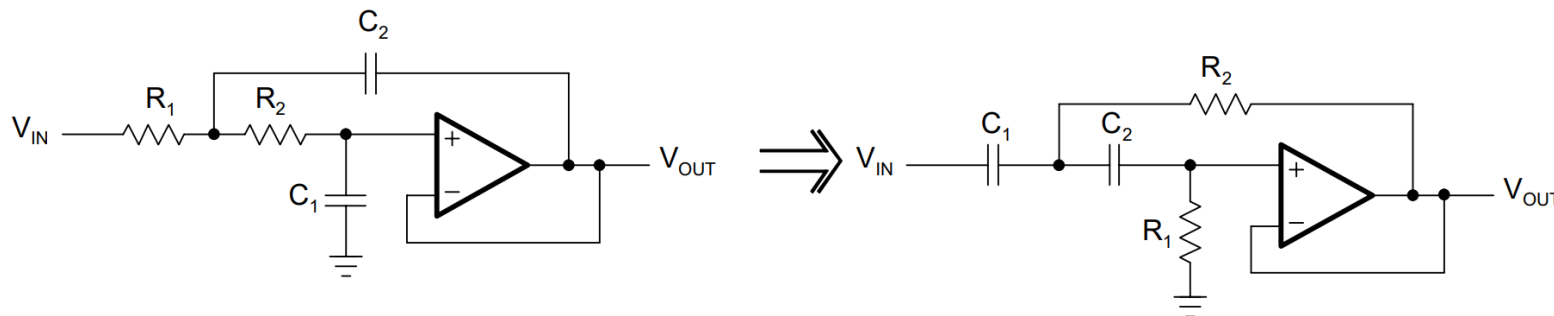


3kHz二阶低通切比雪夫滤波器

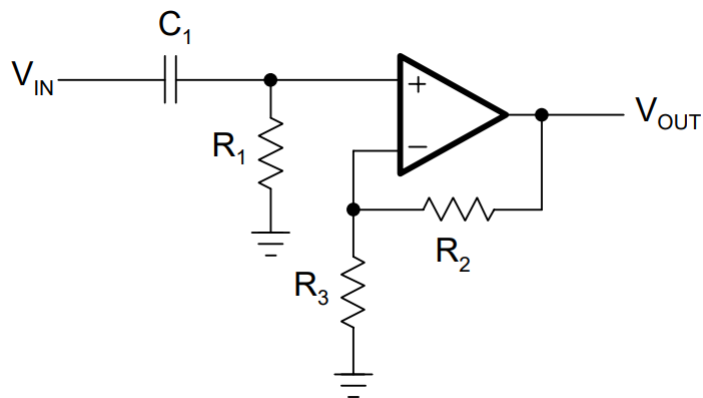
$$A(s) = \frac{A_0}{1 + \omega_c \left[C_1(R_1 + R_2) + (1 - A_0) R_1 C_2 \right] s + \omega_c^2 R_1 R_2 C_1 C_2 s^2}$$

低频滤波器

-----低通转高通滤波器



R/C调换即可



$$A(s) = \frac{A_{\infty}}{\prod_i \left(1 + \frac{a_i}{s} + \frac{b_i}{s^2} \right)}$$

高通一阶滤波器

低频滤波器

-----低通转带通滤波器

$$s \rightarrow \frac{1}{\Delta\Omega} \left(s + \frac{1}{s} \right)$$

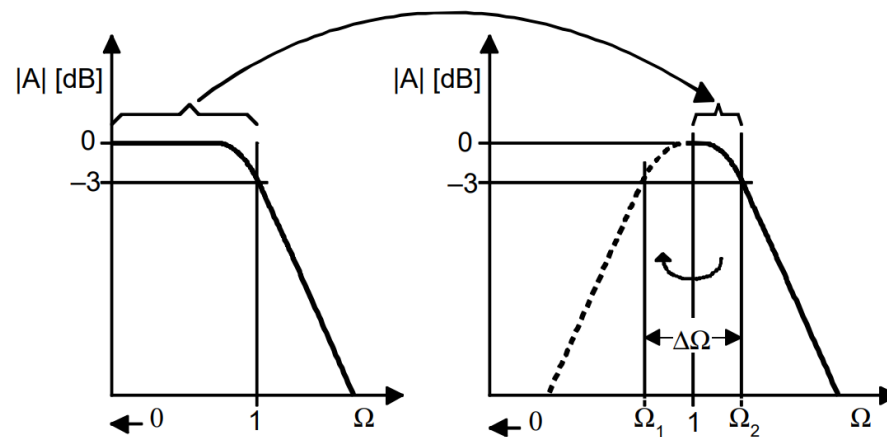
$$\Delta\Omega = \Omega_2 - \Omega_1$$

$$\Omega_m = 1 = \Omega_2 \cdot \Omega_1$$

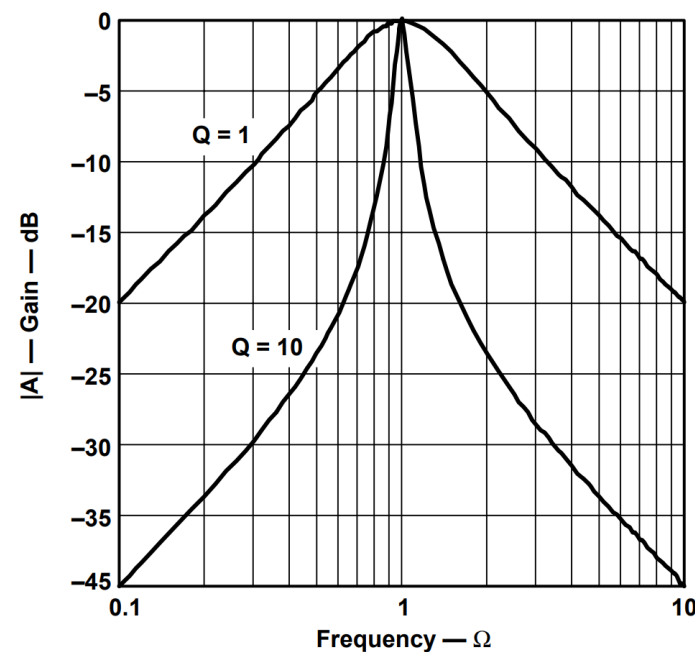
$$Q = \frac{f_m}{B} = \frac{f_m}{f_2 - f_1} = \frac{1}{\Omega_2 - \Omega_1} = \frac{1}{\Delta\Omega}$$

$$A(s) = \frac{A_0}{1 + s} \rightarrow A(s) = \frac{A_0 \cdot \Delta\Omega \cdot s}{1 + \Delta\Omega \cdot s + s^2}$$

$$A(s) = \frac{\frac{A_m}{Q} \cdot s}{1 + \frac{1}{Q} \cdot s + s^2}$$



带通滤波器



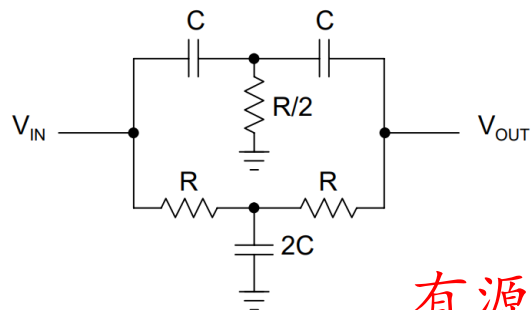
低频滤波器

-----低通转带阻滤波器

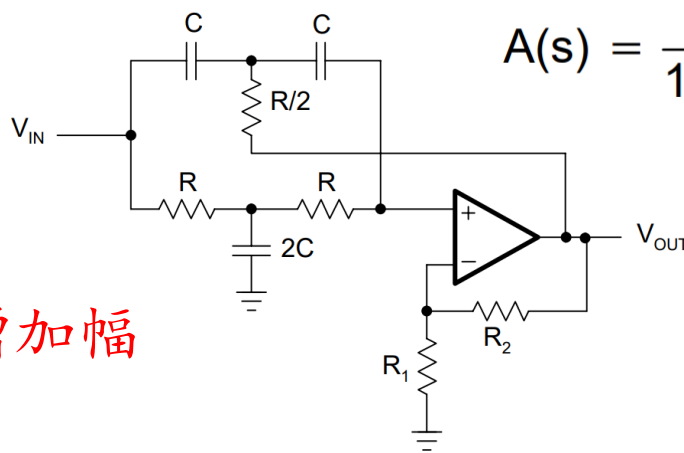
$$s \rightarrow \frac{\Delta\Omega}{s + \frac{1}{s}}$$

$$A(s) = \frac{A_0}{1 + s} \rightarrow A(s) = \frac{A_0(1 + s^2)}{1 + \Delta\Omega \cdot s + s^2}$$

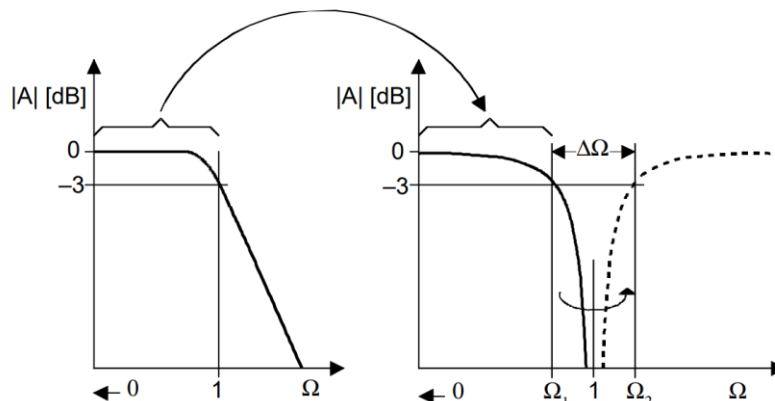
$$A(s) = \frac{A_0(1 + s^2)}{1 + \frac{1}{Q} \cdot s + s^2}$$



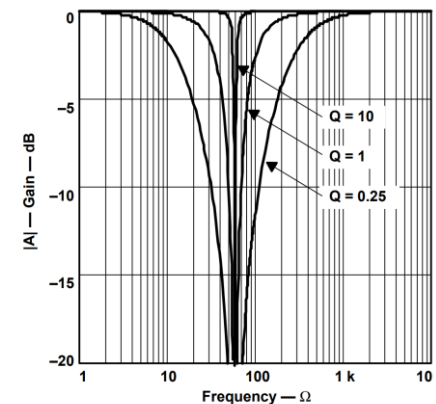
有源可以增加幅
值以及Q值



$$A(s) = \frac{k(1 + s^2)}{1 + 2(2 - k) \cdot s + s^2}$$



带阻滤波器



mid-frequency: $f_m = \frac{1}{2\pi RC}$

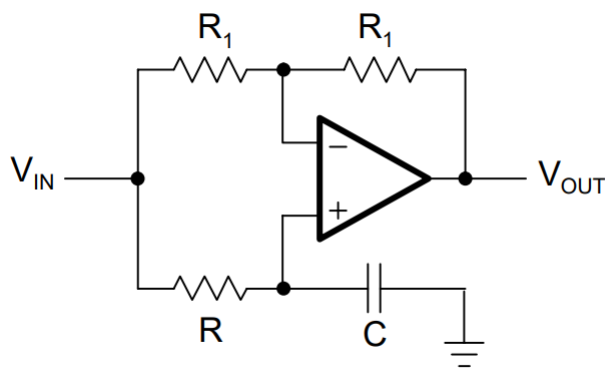
inner gain: $G = 1 + \frac{R_2}{R_1}$

passband gain: $A_0 = G$

rejection quality: $Q = \frac{1}{2(2 - G)}$

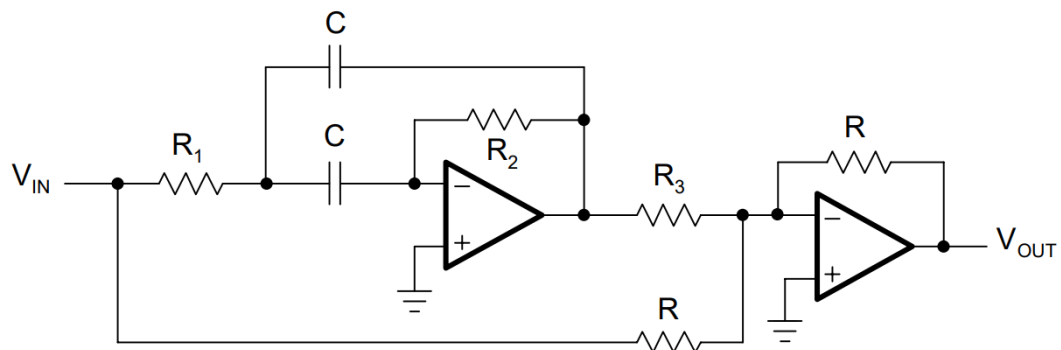
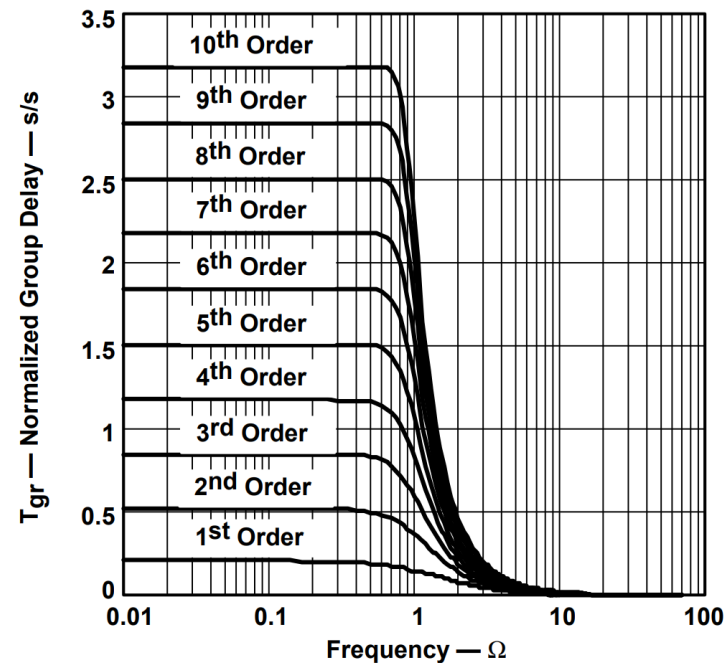
全通滤波器

信号延迟
相位补偿



$$A(s) = \frac{1 - RC\omega_c \cdot s}{1 + RC\omega_c \cdot s}$$

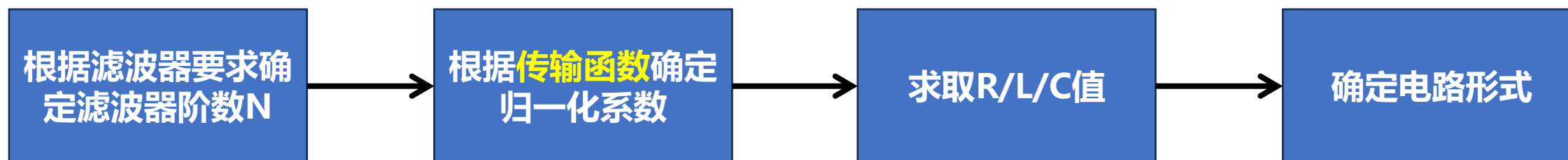
$$t_{gr0} = 2RC$$



$$A(s) = \frac{1 + (2R_1 - \alpha R_2)C\omega_c \cdot s + R_1 R_2 C^2 \omega_c^2 \cdot s^2}{1 + 2R_1 C \omega_c \cdot s + R_1 R_2 C^2 \omega_c^2 \cdot s^2}$$

$$t_{gr0} = 4R_1 C$$

低频滤波器的设计步骤



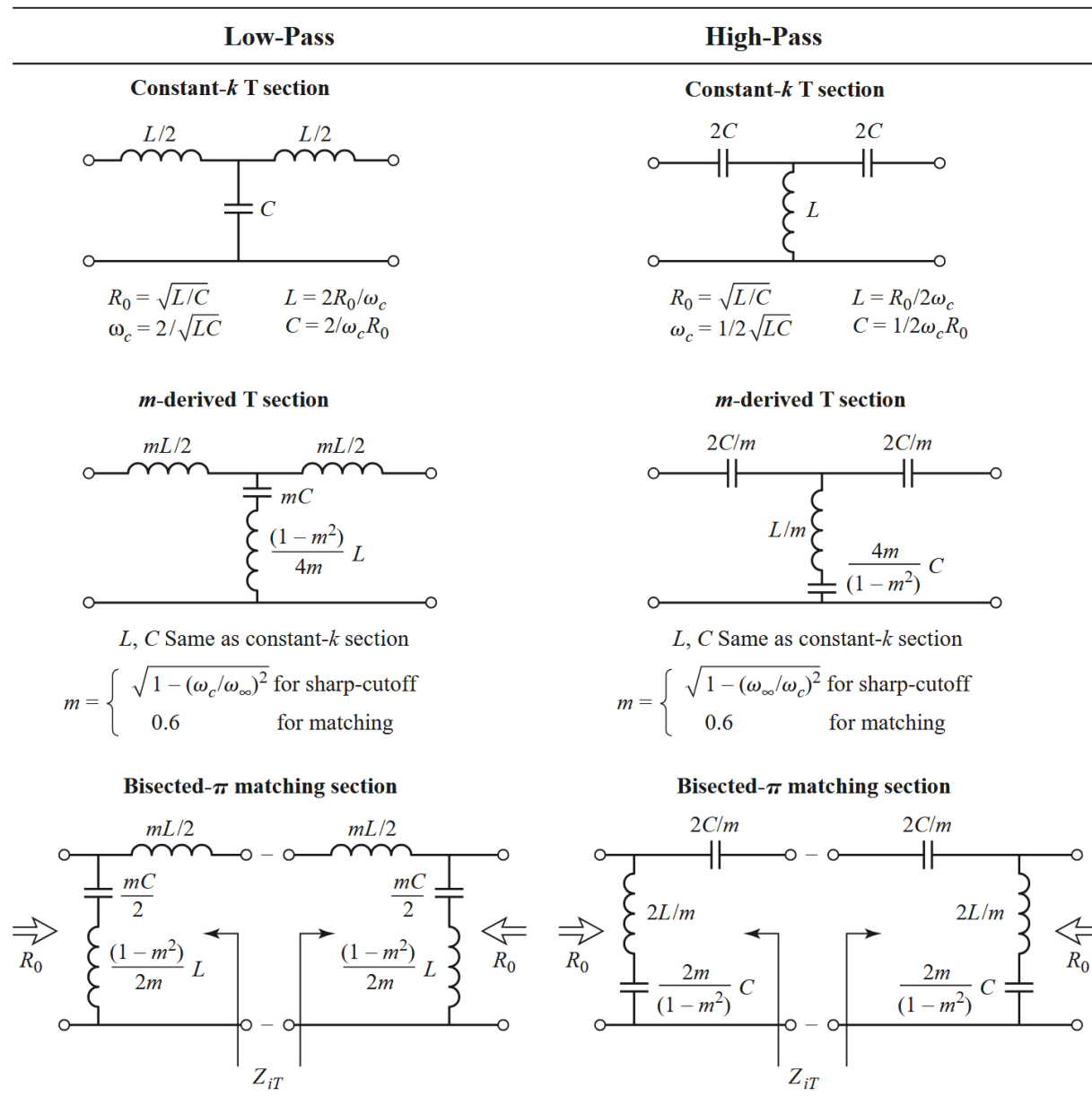
- 电压在射频/微波频段失效，呈现的是波动形式，因此传输函数失效
- 射频/微波出现了波反射现象
- 低频能量主要以电阻热形式损耗
- 射频/微波主要以反射形式“损耗”

与低频滤波器的关键对比

特性	低频滤波器 (音频/电源)	微波滤波器
元件类型	集总RLC, 电阻常被故意加入	分布参数, 尽量避免电阻
阻带能量去向	主要耗散为热 (在电阻/ESR中)	主要反射回源 (几乎不发热)
设计目标	吸收并消耗阻带能量 (如电源去耦)	反射阻带能量, 保护通带信号
发热情况	显著, 需考虑功率容量	极微小, 通常可忽略
典型例子	音频分频器 (电阻发热)	微带带通滤波器 (阻带驻波很高)

射频/微波滤波器

- 镜像参数法（定k法，m推演法）——频率响应无法精确指定
- 插入损耗法——设计方式更自由，设计流程更规范



插入损耗法

$$P_{LR}(\text{插入损耗}) = \frac{\text{源提供的功率}}{\text{传输至负载的功率}} = \frac{P_{inc}}{P_{load}} = \frac{1}{1 - |\Gamma(\omega)|^2}$$

$$v(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} V(\omega) e^{j\omega t} d\omega, \xrightarrow{v(t) = v^*(t)} \int_{-\infty}^{\infty} V(\omega) e^{j\omega t} d\omega = \int_{-\infty}^{\infty} V^*(\omega) e^{-j\omega t} d\omega = \int_{-\infty}^{\infty} V^*(-\omega) e^{j\omega t} d\omega, \Rightarrow V(-\omega) = V^*(\omega),$$

$Re(V(\omega))$ 为频率域偶函数, $Im(V(\omega))$ 为频率域奇函数

$$V^*(-\omega) = Z^*(-\omega)I^*(-\omega) = Z^*(-\omega)I(\omega) = V(\omega) = Z(\omega)I(\omega).$$

$Z(\omega) = R(\omega) + jX(\omega)$, $R(\omega)$ 为偶函数, $X(\omega)$ 为奇函数

$$\Gamma(\omega) = \frac{Z(\omega) - Z_0}{Z(\omega) + Z_0} = \frac{R(\omega) - Z_0 + jX(\omega)}{R(\omega) + Z_0 + jX(\omega)}, \Rightarrow \Gamma(-\omega) = \frac{R(\omega) - Z_0 - jX(\omega)}{R(\omega) + Z_0 - jX(\omega)} = \Gamma^*(\omega)$$

$Re(\Gamma(\omega))$ 为频率域偶函数, $Im(\Gamma(\omega))$ 为频率域奇函数

$$|\Gamma(\omega)|^2 = \Gamma(\omega)\Gamma^*(\omega) = \Gamma(\omega)\Gamma(-\omega) = |\Gamma(-\omega)|^2 \quad |\Gamma(\omega)| \text{ 频率域偶函数} \quad |\Gamma(\omega)|^2 = \frac{M(\omega^2)}{M(\omega^2) + N(\omega^2)}$$

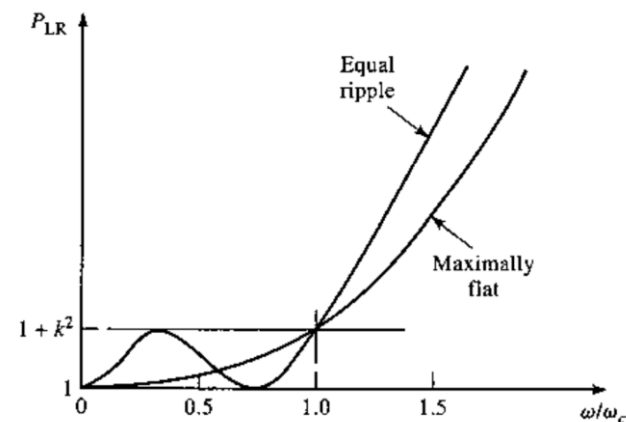
插入损耗法

$$P_{LR} = 1 + \frac{M(\omega^2)}{N(\omega^2)}.$$

□ 巴特沃斯型滤波器（二项式型）

$$P_{LR} = 1 + k^2 \left(\frac{\omega}{\omega_c} \right)^{2N}$$

N 为滤波器阶数， ω 为截止频率，-3 dB作为截止频率，则 $k=1$ ，当
 $\omega \gg \omega_c$ 时， $P_{LR} \approx k^2 \left(\frac{\omega}{\omega_c} \right)^{2N}$ ，这表明插入损耗以 $20N \text{ dB/十倍频程}$ 的速率增加。



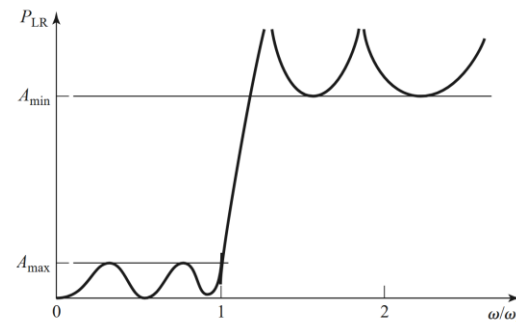
□ 切比雪夫型滤波器（等波纹型）

$$P_{LR} = 1 + k^2 T_N^2 \left(\frac{\omega}{\omega_c} \right)$$

□ 椭圆型滤波器

□ 线性相位滤波器

$$\phi(\omega) = A\omega \left[1 + p \left(\frac{\omega}{\omega_c} \right)^{2N} \right],$$



巴特沃斯（最大平坦）型滤波器

$$P_{LR} = 1 + \omega^4.$$

$$Z_{in} = j\omega L + \frac{R(1 - j\omega RC)}{1 + \omega^2 R^2 C^2}, \quad \Gamma = \frac{Z_{in} - 1}{Z_{in} + 1},$$

$$P_{LR} = \frac{1}{1 - |\Gamma|^2} = \frac{1}{1 - [(Z_{in} - 1)/(Z_{in} + 1)][(Z_{in}^* - 1)/(Z_{in}^* + 1)]} = \frac{|Z_{in} + 1|^2}{2(Z_{in} + Z_{in}^*)}.$$

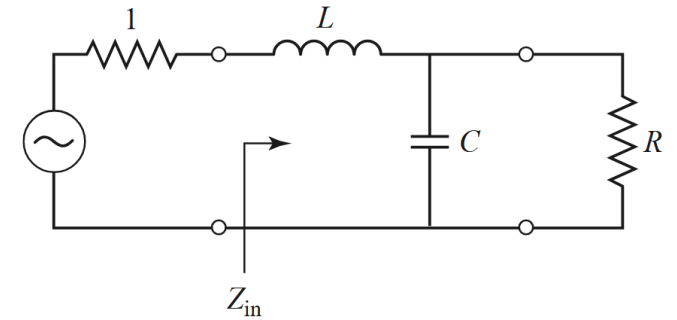
$$Z_{in} + Z_{in}^* = \frac{2R}{1 + \omega^2 R^2 C^2},$$

$$|Z_{in} + 1|^2 = \left(\frac{R}{1 + \omega^2 R^2 C^2} + 1 \right)^2 + \left(\omega L - \frac{\omega C R^2}{1 + \omega^2 R^2 C^2} \right)^2,$$

$$P_{LR} = \frac{1 + \omega^2 R^2 C^2}{4R} \left[\left(\frac{R}{1 + \omega^2 R^2 C^2} + 1 \right)^2 + \left(\omega L - \frac{\omega C R^2}{1 + \omega^2 R^2 C^2} \right)^2 \right]$$

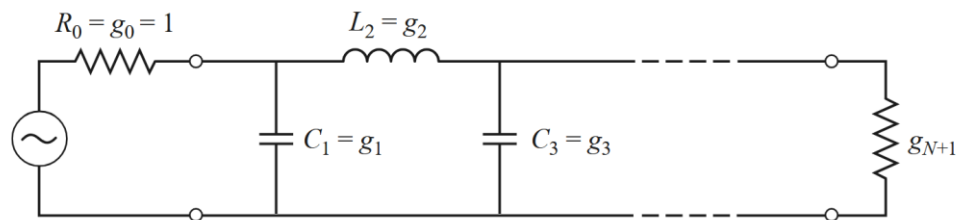
$$= \frac{1}{4R} (R^2 + 2R + 1 + R^2 \omega^2 C^2 + \omega^2 L^2 + \omega^4 L^2 C^2 R^2 - 2\omega^2 L C R^2)$$

$$= 1 + \frac{1}{4R} [(1 - R)^2 + (R^2 C^2 + L^2 - 2LCR^2)\omega^2 + L^2 C^2 R^2 \omega^4].$$

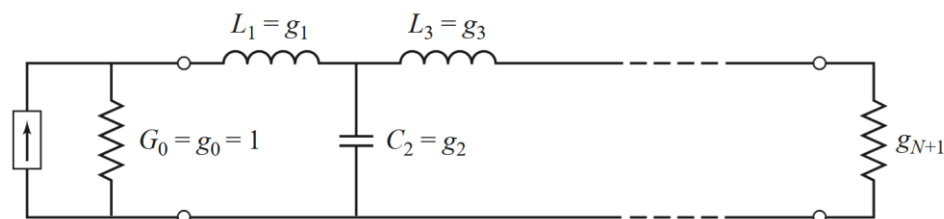


$$\Rightarrow L = C = \sqrt{2}.$$

巴特沃斯型滤波器设计表格



(a)



(b)

TABLE 8.3 Element Values for Maximally Flat Low-Pass Filter Prototypes ($g_0 = 1$, $\omega_c = 1$, $N = 1$ to 10)

N	g_1	g_2	g_3	g_4	g_5	g_6	g_7	g_8	g_9	g_{10}	g_{11}
1	2.0000	1.0000									
2	1.4142	1.4142	1.0000								
3	1.0000	2.0000	1.0000	1.0000							
4	0.7654	1.8478	1.8478	0.7654	1.0000						
5	0.6180	1.6180	2.0000	1.6180	0.6180	1.0000					
6	0.5176	1.4142	1.9318	1.9318	1.4142	0.5176	1.0000				
7	0.4450	1.2470	1.8019	2.0000	1.8019	1.2470	0.4450	1.0000			
8	0.3902	1.1111	1.6629	1.9615	1.9615	1.6629	1.1111	0.3902	1.0000		
9	0.3473	1.0000	1.5321	1.8794	2.0000	1.8794	1.5321	1.0000	0.3473	1.0000	
10	0.3129	0.9080	1.4142	1.7820	1.9754	1.9754	1.7820	1.4142	0.9080	0.3129	1.0000

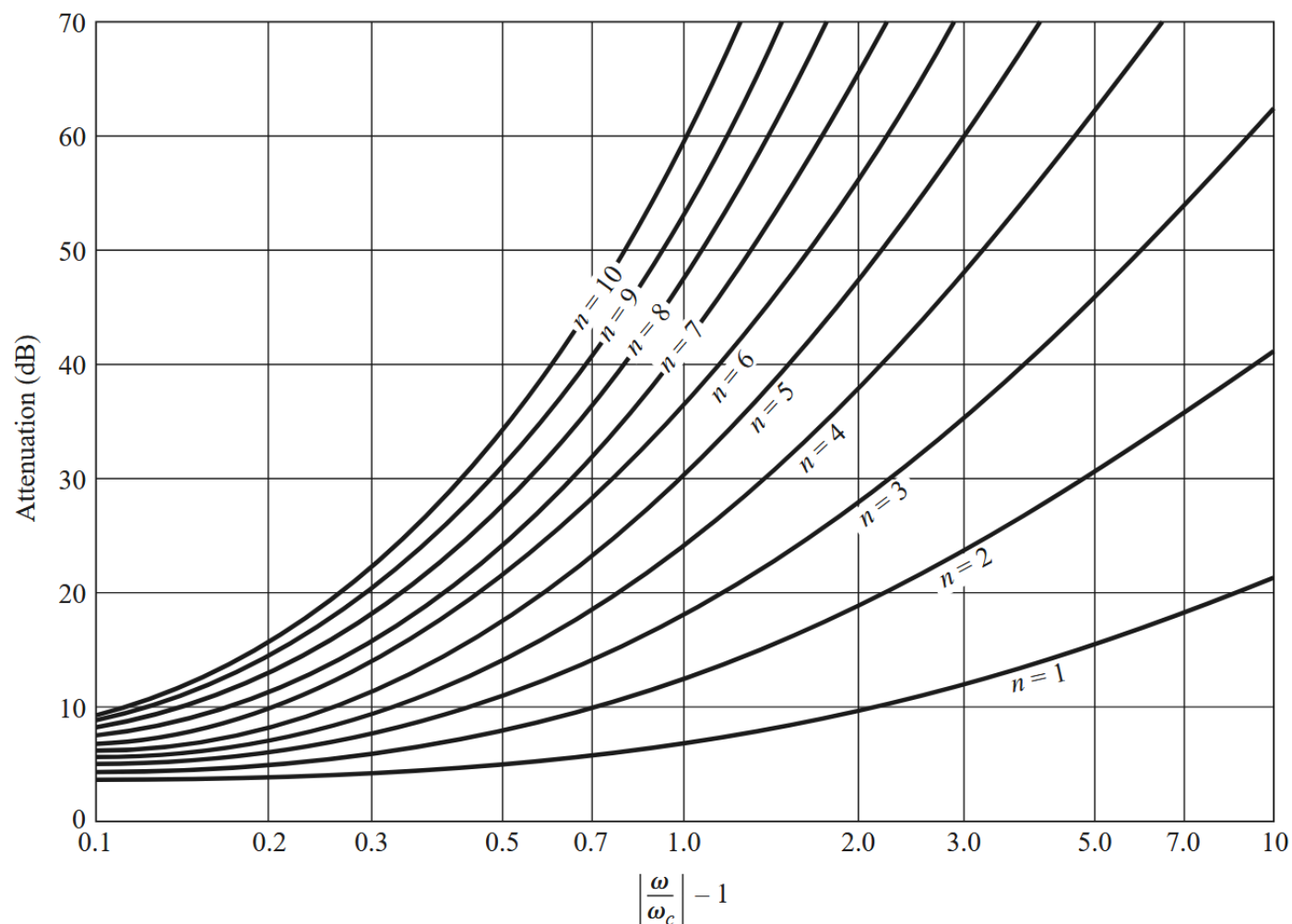
Source: Reprinted from G. L. Matthaei, L. Young, and E. M. T. Jones, *Microwave Filters, Impedance-Matching Networks, and Coupling Structures*, Artech House, Dedham, Mass., 1980, with permission.

$$g_0 = \begin{cases} \text{generator resistance (network of Figure 8.25a)} \\ \text{generator conductance (network of Figure 8.25b)} \end{cases}$$

$$g_k \quad (k = 1 \text{ to } N) = \begin{cases} \text{inductance for series inductors} \\ \text{capacitance for shunt capacitors} \end{cases}$$

$$g_{N+1} = \begin{cases} \text{load resistance if } g_N \text{ is a shunt capacitor} \\ \text{load conductance if } g_N \text{ is a series inductor} \end{cases}$$

巴特沃斯型滤波器



最大平坦滤波器的衰减与归一化频率关系。

切比雪夫型滤波器

$$P_{LR} = 1 + k^2 T_N^2(\omega), \quad T_N \text{ 为 } N \text{ 阶切比雪夫多项式}$$

$$P_{LR} = \frac{1 + \omega^2 R^2 C^2}{4R} \left[\left(\frac{R}{1 + \omega^2 R^2 C^2} + 1 \right)^2 + \left(\omega L - \frac{\omega C R^2}{1 + \omega^2 R^2 C^2} \right)^2 \right]$$

$$= \frac{1}{4R} (R^2 + 2R + 1 + R^2 \omega^2 C^2 + \omega^2 L^2 + \omega^4 L^2 C^2 R^2 - 2\omega^2 L C R^2)$$

$$= 1 + \frac{1}{4R} [(1 - R)^2 + (R^2 C^2 + L^2 - 2LCR^2)\omega^2 + L^2 C^2 R^2 \omega^4].$$

$$1 + k^2 (4\omega^4 - 4\omega^2 + 1) = 1 + \frac{1}{4R} [(1 - R)^2 + (R^2 C^2 + L^2 - 2LCR^2)\omega^2 + L^2 C^2 R^2 \omega^4],$$

$$k^2 = \frac{(1 - R)^2}{4R},$$

$$4k^2 = \frac{1}{4R} L^2 C^2 R^2,$$

$$-4k^2 = \frac{1}{4R} (R^2 C^2 + L^2 - 2LCR^2),$$

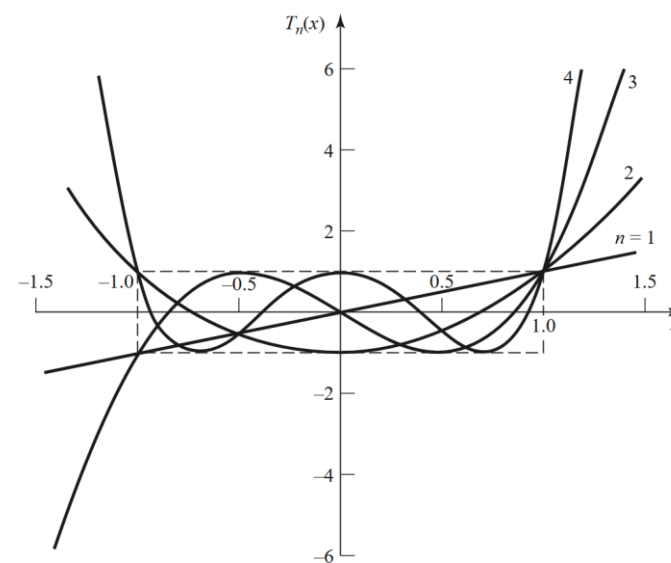
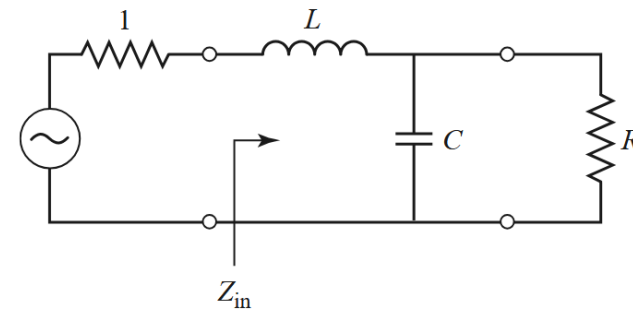
$$T_1(x) = x,$$

$$T_2(x) = 2x^2 - 1,$$

$$T_3(x) = 4x^3 - 3x,$$

$$T_4(x) = 8x^4 - 8x^2 + 1.$$

$$T_n(x) = 2x T_{n-1}(x) - T_{n-2}(x).$$



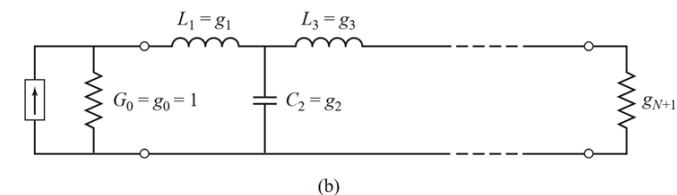
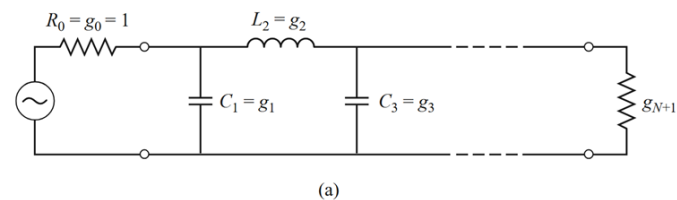
切比雪夫型滤波器设计表格

0.5 dB Ripple											
N	g_1	g_2	g_3	g_4	g_5	g_6	g_7	g_8	g_9	g_{10}	g_{11}
1	0.6986	1.0000									
2	1.4029	0.7071	1.9841								
3	1.5963	1.0967	1.5963	1.0000							
4	1.6703	1.1926	2.3661	0.8419	1.9841						
5	1.7058	1.2296	2.5408	1.2296	1.7058	1.0000					
6	1.7254	1.2479	2.6064	1.3137	2.4758	0.8696	1.9841				
7	1.7372	1.2583	2.6381	1.3444	2.6381	1.2583	1.7372	1.0000			
8	1.7451	1.2647	2.6564	1.3590	2.6964	1.3389	2.5093	0.8796	1.9841		
9	1.7504	1.2690	2.6678	1.3673	2.7239	1.3673	2.6678	1.2690	1.7504	1.0000	
10	1.7543	1.2721	2.6754	1.3725	2.7392	1.3806	2.7231	1.3485	2.5239	0.8842	1.9841

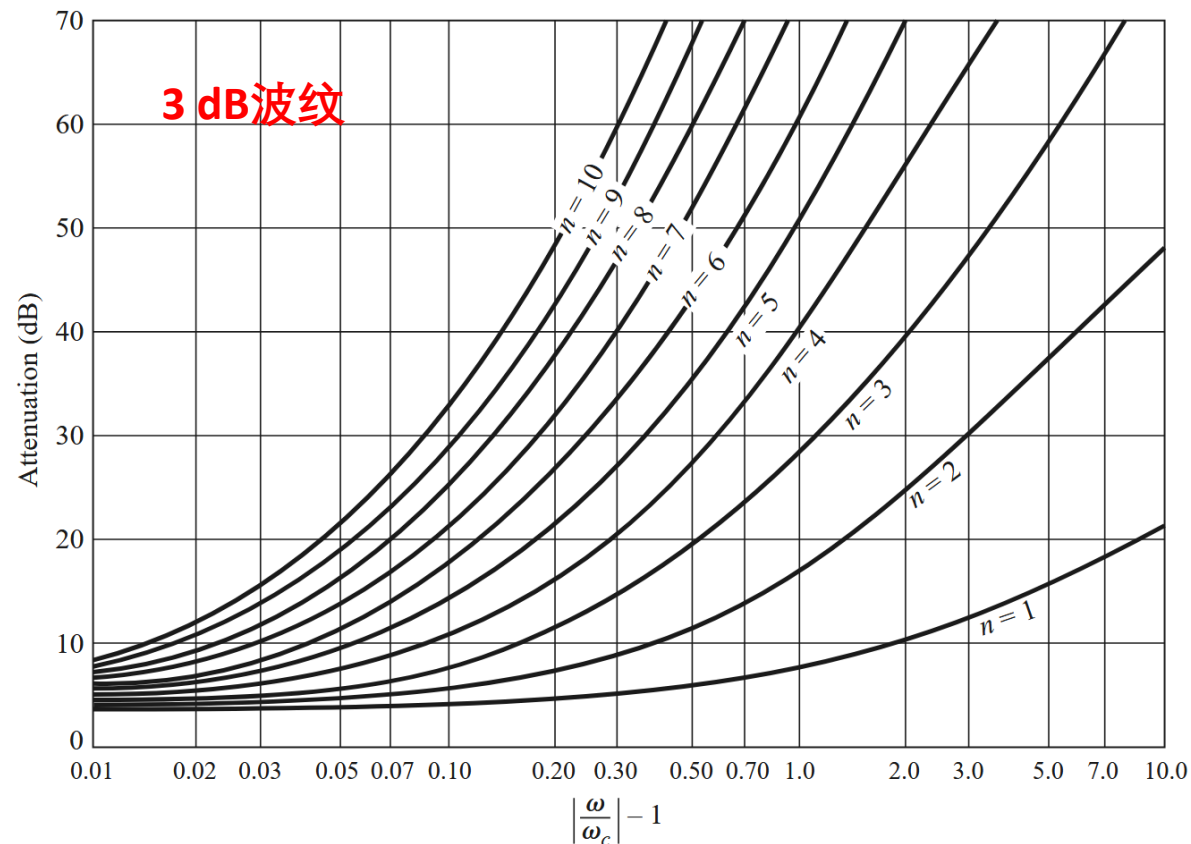
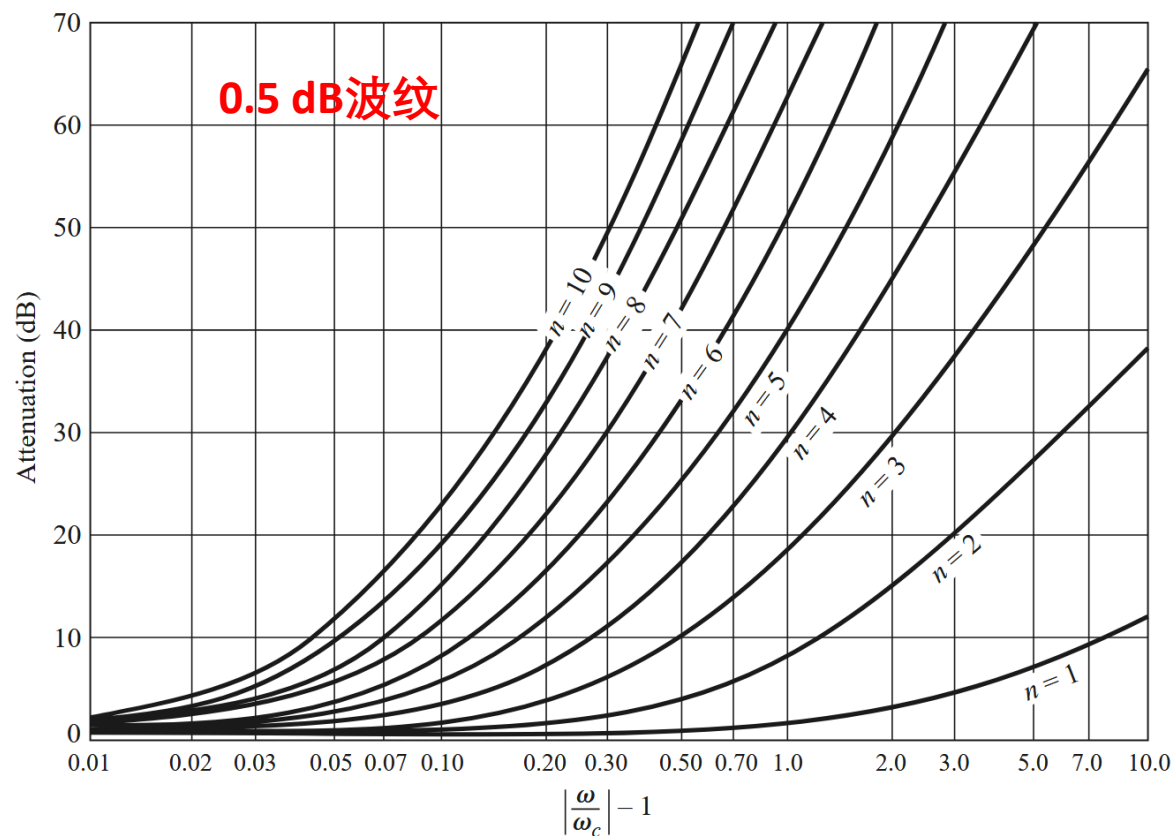
3.0 dB Ripple											
N	g_1	g_2	g_3	g_4	g_5	g_6	g_7	g_8	g_9	g_{10}	g_{11}
1	1.9953	1.0000									
2	3.1013	0.5339	5.8095								
3	3.3487	0.7117	3.3487	1.0000							
4	3.4389	0.7483	4.3471	0.5920	5.8095						
5	3.4817	0.7618	4.5381	0.7618	3.4817	1.0000					
6	3.5045	0.7685	4.6061	0.7929	4.4641	0.6033	5.8095				
7	3.5182	0.7723	4.6386	0.8039	4.6386	0.7723	3.5182	1.0000			
8	3.5277	0.7745	4.6575	0.8089	4.6990	0.8018	4.4990	0.6073	5.8095		
9	3.5340	0.7760	4.6692	0.8118	4.7272	0.8118	4.6692	0.7760	3.5340	1.0000	
10	3.5384	0.7771	4.6768	0.8136	4.7425	0.8164	4.7260	0.8051	4.5142	0.6091	5.8095

Source: Reprinted from G. L. Matthaei, L. Young, and E. M. T. Jones, *Microwave Filters, Impedance-Matching Networks, and Coupling Structures*, Artech House, Dedham, Mass., 1980, with permission.

□ 对于 N 为偶数时，负载电阻不再是归一化电阻1，解决方法：阻抗匹配或者使用奇数阶



切比雪夫型滤波器



Attenuation versus normalized frequency for equal-ripple filter prototypes.
(a) 0.5 dB ripple level. (b) 3.0 dB ripple level.

Adapted from G. L. Matthaei, L. Young, and E. M. T. Jones, *Microwave Filters, Impedance-Matching Networks, and Coupling Structures*, Artech House, Dedham, Mass., 1980, with permission.

滤波器的衰减与归一化频率关系。

相位最平坦（贝塞尔型）滤波器设计表格

TABLE 8.5 Element Values for Maximally Flat Time Delay Low-Pass Filter Prototypes
($g_0 = 1, \omega_c = 1, N = 1$ to 10)

N	g_1	g_2	g_3	g_4	g_5	g_6	g_7	g_8	g_9	g_{10}	g_{11}
1	2.0000	1.0000									
2	1.5774	0.4226	1.0000								
3	1.2550	0.5528	0.1922	1.0000							
4	1.0598	0.5116	0.3181	0.1104	1.0000						
5	0.9303	0.4577	0.3312	0.2090	0.0718	1.0000					
6	0.8377	0.4116	0.3158	0.2364	0.1480	0.0505	1.0000				
7	0.7677	0.3744	0.2944	0.2378	0.1778	0.1104	0.0375	1.0000			
8	0.7125	0.3446	0.2735	0.2297	0.1867	0.1387	0.0855	0.0289	1.0000		
9	0.6678	0.3203	0.2547	0.2184	0.1859	0.1506	0.1111	0.0682	0.0230	1.0000	
10	0.6305	0.3002	0.2384	0.2066	0.1808	0.1539	0.1240	0.0911	0.0557	0.0187	1.0000

Source: Reprinted from G. L. Matthaei, L. Young, and E. M. T. Jones, *Microwave Filters, Impedance-Matching Networks, and Coupling Structures*, Artech House, Dedham, Mass., 1980, with permission.

反归一化

$$P'_{\text{LR}}(\omega) = P_{\text{LR}}\left(\frac{\omega}{\omega_c}\right),$$

$$jX_k = j\frac{\omega}{\omega_c}L_k = j\omega L'_k,$$

$$jB_k = j\frac{\omega}{\omega_c}C_k = j\omega C'_k,$$



$$L'_k = \frac{L_k}{\omega_c},$$

$$C'_k = \frac{C_k}{\omega_c}.$$

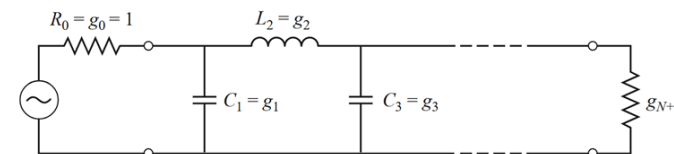


$$L'_k = \frac{R_0 L_k}{\omega_c},$$

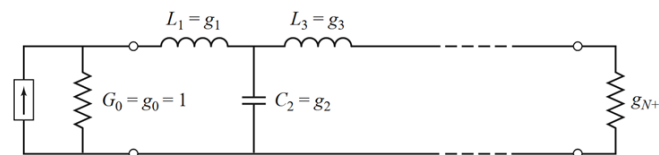
$$C'_k = \frac{C_k}{R_0 \omega_c}.$$

反归一化频率

反归一化电阻



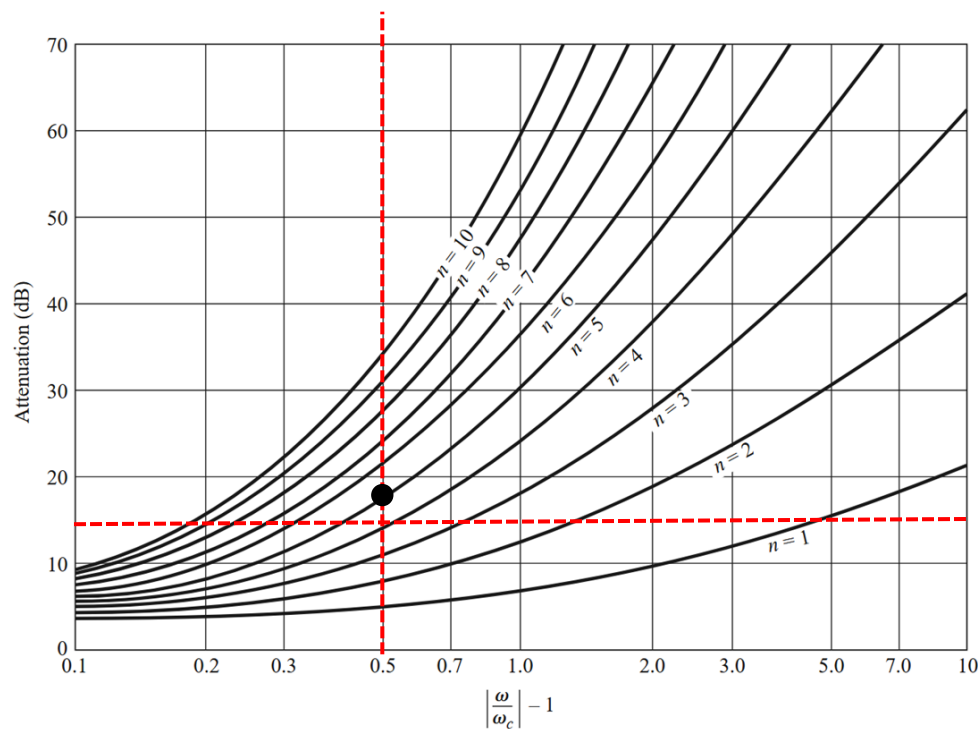
(a)



(b)

例题

设计一个最大平坦低通滤波器，其截止频率为 2 GHz，特性阻抗为 50 Ω，且在 3 GHz 处具有至少 15 dB 的插入损耗。计算并绘制在频率范围 0 到 4 GHz 内的幅频响应和群延迟，并将结果与相同阶数的等波纹（3.0 dB 波纹）和线性相位滤波器进行比较。



$$\left| \frac{\omega}{\omega_c} \right| - 1 = 0.5 \quad N = 5$$

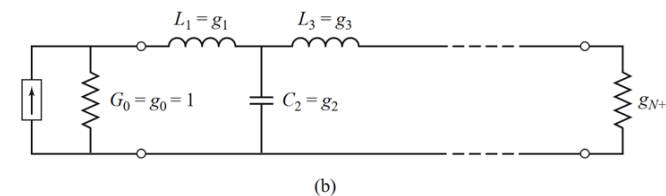
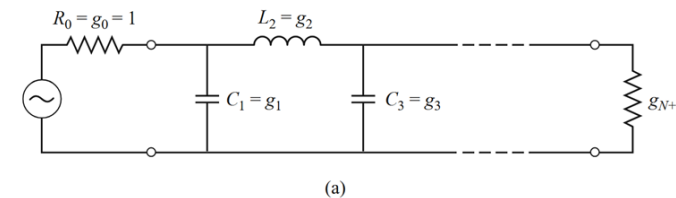


TABLE 8.3 Element Values for Maximally Flat Low-Pass Filter Prototypes ($g_0 = 1$, $\omega_c = 1$, $N = 1$ to 10)

N	g ₁	g ₂	g ₃	g ₄	g ₅	g ₆	g ₇	g ₈	g ₉	g ₁₀	g ₁₁
1	2.0000	1.0000									
2	1.4142	1.4142	1.0000								
3	1.0000	2.0000	1.0000	1.0000							
4	0.7654	1.8478	1.8478	0.7654	1.0000						
5	0.6180	1.6180	2.0000	1.6180	0.6180	1.0000					
6	0.5176	1.4142	1.9318	1.9318	1.4142	0.5176	1.0000				
7	0.4450	1.2470	1.8019	2.0000	1.8019	1.2470	0.4450	1.0000			
8	0.3902	1.1111	1.6629	1.9615	1.9615	1.6629	1.1111	0.3902	1.0000		
9	0.3473	1.0000	1.5321	1.8794	2.0000	1.8794	1.5321	1.0000	0.3473	1.0000	
10	0.3129	0.9080	1.4142	1.7820	1.9754	1.9754	1.7820	1.4142	0.9080	0.3129	1.0000

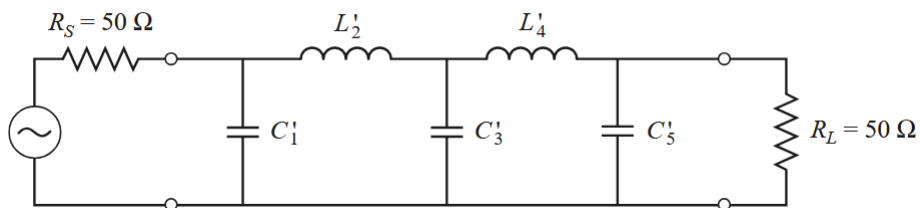
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$$L'_k = \frac{L_k R_0}{2\pi f_c}$$

$$C'_k = \frac{C_k}{R_0 2\pi f_c}$$

例题

设计一个最大平坦低通滤波器，其截止频率为 2 GHz，特性阻抗为 50 Ω，且在 3 GHz 处具有至少 15 dB 的插入损耗。计算并绘制在频率范围 0 到 4 GHz 内的幅频响应和群延迟，并将结果与相同阶数的等波纹（3.0 dB 波纹）和线性相位滤波器进行比较。



$$\begin{aligned} C'_1 &= 0.984 \text{ pF}, \\ L'_2 &= 6.438 \text{ nH}, \\ C'_3 &= 3.183 \text{ pF}, \\ L'_4 &= 6.438 \text{ nH}, \\ C'_5 &= 0.984 \text{ pF}. \end{aligned}$$

3.0 dB Ripple											
N	g_1	g_2	g_3	g_4	g_5	g_6	g_7	g_8	g_9	g_{10}	g_{11}
1	1.9953	1.0000									
2	3.1013	0.5339	5.8095								
3	3.3487	0.7117	3.3487	1.0000							
4	3.4389	0.7483	4.3471	0.5920	5.8095						
5	3.4817	0.7618	4.5381	0.7618	3.4817	1.0000					
6	3.5045	0.7685	4.6061	0.7929	4.4641	0.6033	5.8095				
7	3.5182	0.7723	4.6386	0.8039	4.6386	0.7723	3.5182	1.0000			
8	3.5277	0.7745	4.6575	0.8089	4.6990	0.8018	4.4990	0.6073	5.8095		
9	3.5340	0.7760	4.6692	0.8118	4.7272	0.8118	4.6692	0.7760	3.5340	1.0000	
10	3.5384	0.7771	4.6768	0.8136	4.7425	0.8164	4.7260	0.8051	4.5142	0.6091	5.8095

切比雪夫型

切比雪夫型

$$\begin{aligned} C'_1 &= 5.541 \text{ pF} \\ L'_2 &= 3.031 \text{ nH} \\ C'_3 &= 7.212 \text{ pF} \\ L'_4 &= 3.031 \text{ nH} \\ C'_5 &= 5.541 \text{ pF} \end{aligned}$$

Source: Reprinted from G. L. Matthaei, L. Young, and E. M. T. Jones, *Microwave Filters, Impedance-Matching Networks, and Coupling Structures*, Artech House, Dedham, Mass., 1980, with permission.

TABLE 8.5 Element Values for Maximally Flat Time Delay Low-Pass Filter Prototypes ($g_0 = 1, \omega_c = 1, N = 1$ to 10)

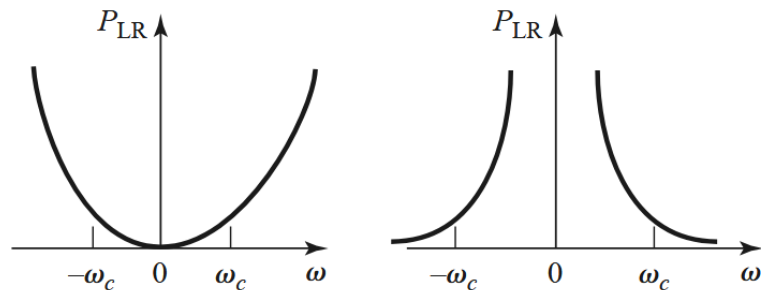
N	g_1	g_2	g_3	g_4	g_5	g_6	g_7	g_8	g_9	g_{10}	g_{11}
1	2.0000	1.0000									
2	1.5774	0.4226	1.0000								
3	1.2550	0.5528	0.1922	1.0000							
4	1.0598	0.5116	0.3181	0.1104	1.0000						
5	0.9303	0.4577	0.3312	0.2090	0.0718	1.0000					
6	0.8377	0.4116	0.3158	0.2364	0.1480	0.0505	1.0000				
7	0.7677	0.3744	0.2944	0.2378	0.1778	0.1104	0.0375	1.0000			
8	0.7125	0.3446	0.2735	0.2297	0.1867	0.1387	0.0855	0.0289	1.0000		
9	0.6678	0.3203	0.2547	0.2184	0.1859	0.1506	0.1111	0.0682	0.0230	1.0000	
10	0.6305	0.3002	0.2384	0.2066	0.1808	0.1539	0.1240	0.0911	0.0557	0.0187	1.0000

相位最大平坦

$$\begin{aligned} C'_1 &= 1.481 \text{ pF} \\ L'_2 &= 1.821 \text{ nH} \\ C'_3 &= 0.527 \text{ pF} \\ L'_4 &= 0.832 \text{ nH} \\ C'_5 &= 0.114 \text{ pF} \end{aligned}$$

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高通滤波器



$$\frac{\omega_c}{\omega} \rightarrow \omega_{normalized}$$



此时，电感和电容相应
转化为容抗和感抗



为保证电感和电容均为
正值，需加负号

例如： $j\omega L \rightarrow \frac{j1}{\omega} L \rightarrow \frac{1}{-j\omega 1/L}$

$$-\frac{\omega_c}{\omega} \rightarrow \omega_{normalized}$$

低通串联电抗转化为 $jX_k = -j\frac{\omega_c}{\omega} L_k = \frac{1}{j\omega C'_k}$,

低通并联电纳转化为 $jB_k = -j\frac{\omega_c}{\omega} C_k = \frac{1}{j\omega L'_k}$,



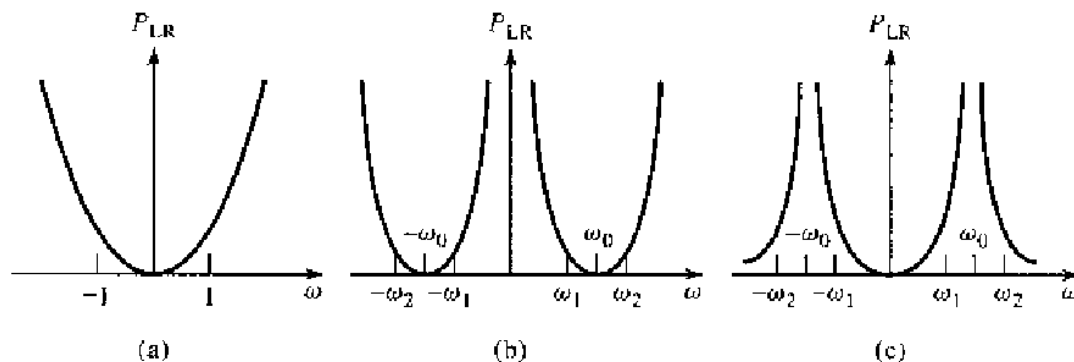
反归一化电阻

$$C'_k = \frac{1}{R_0 \omega_c L_k},$$

$$L'_k = \frac{R_0}{\omega_c C_k}.$$

低通串联电感转化为串联电容；
低通并联电容转化为并联电感

带通滤波器



低通

带通

带阻

$$\omega \leftarrow \frac{\omega_0}{\omega_2 - \omega_1} \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right) = \frac{1}{\Delta} \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right), \quad \omega_0 = \sqrt{\omega_1 \omega_2}.$$

$$\Delta = \frac{\omega_2 - \omega_1}{\omega_0}$$

近似等效
方便计算

When $\omega = \omega_0$, $\frac{1}{\Delta} \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right) = 0$.

When $\omega = \omega_1$, $\frac{1}{\Delta} \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right) = \frac{1}{\Delta} \left(\frac{\omega_1^2 - \omega_0^2}{\omega_0 \omega_1} \right) = -1$.

When $\omega = \omega_2$, $\frac{1}{\Delta} \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right) = \frac{1}{\Delta} \left(\frac{\omega_2^2 - \omega_0^2}{\omega_0 \omega_2} \right) = 1$.

低通串联电抗经过频率映射后变为:

$$jX_k = \frac{j}{\Delta} \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right) L_k = j \frac{\omega L_k}{\Delta \omega_0} - j \frac{\omega_0 L_k}{\Delta \omega} = j\omega L'_k - j \frac{1}{\omega C'_k},$$

低通串联电感变成了电感与
电容的串联

$$L'_k = \frac{L_k}{\Delta \omega_0},$$

$$C'_k = \frac{\Delta}{\omega_0 L_k}.$$

反归一化
电阻

$$L'_k * R_0$$

低通并联电纳经过频率映射后变为:

$$jB_k = \frac{j}{\Delta} \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right) C_k = j \frac{\omega C_k}{\Delta \omega_0} - j \frac{\omega_0 C_k}{\Delta \omega} = j\omega C'_k - j \frac{1}{\omega L'_k},$$

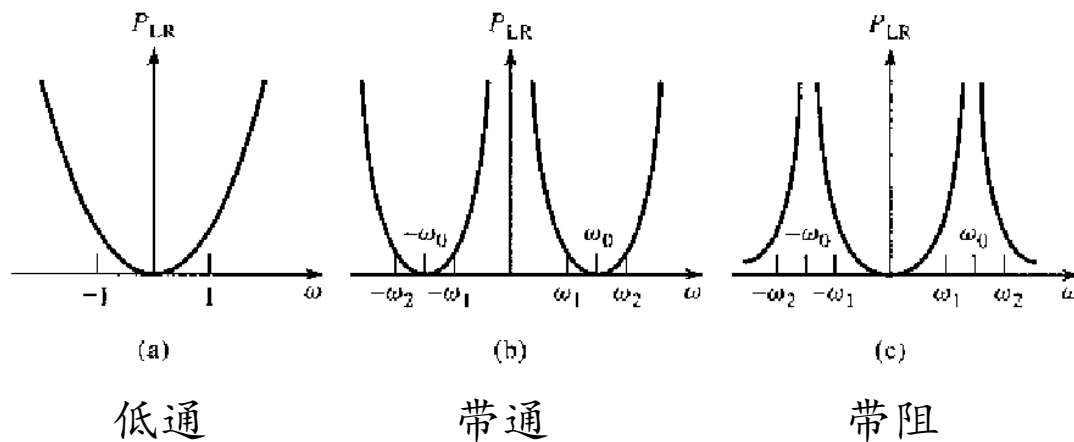
低通并联电容变成了电感与
电容的并联

$$L'_k = \frac{\Delta}{\omega_0 C_k},$$

$$C'_k = \frac{C_k}{\Delta \omega_0}.$$

$$\frac{C'_k}{R_0}$$

带阻滤波器



$$\omega \leftarrow -\Delta \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right)^{-1}$$

低通串联电感变成了电感与
电容的**并联**

$$L'_k = \frac{\Delta L_k}{\omega_0},$$

$$C'_k = \frac{1}{\omega_0 \Delta L_k}.$$

低通并联电容变成了电感与
电容的**串联**

$$L'_k = \frac{1}{\omega_0 \Delta C_k},$$

$$C'_k = \frac{\Delta C_k}{\omega_0}.$$

反归一化
电阻

$$L'_k * R_0$$

$$\frac{C'_k}{R_0}$$

总结

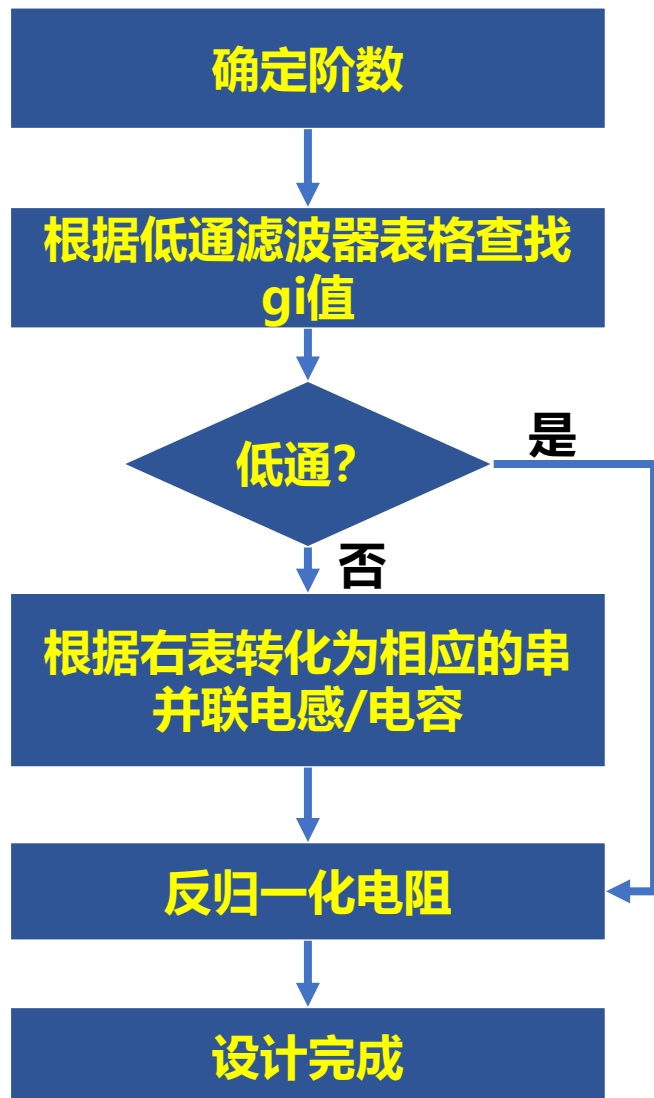
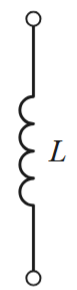
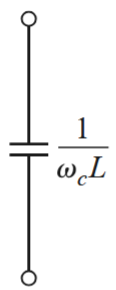
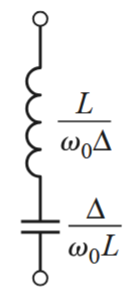
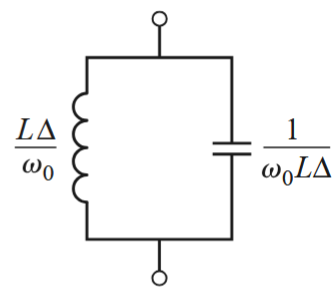
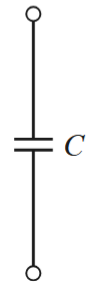
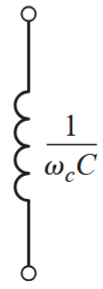
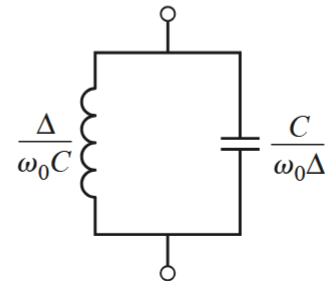
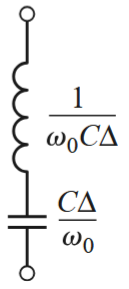
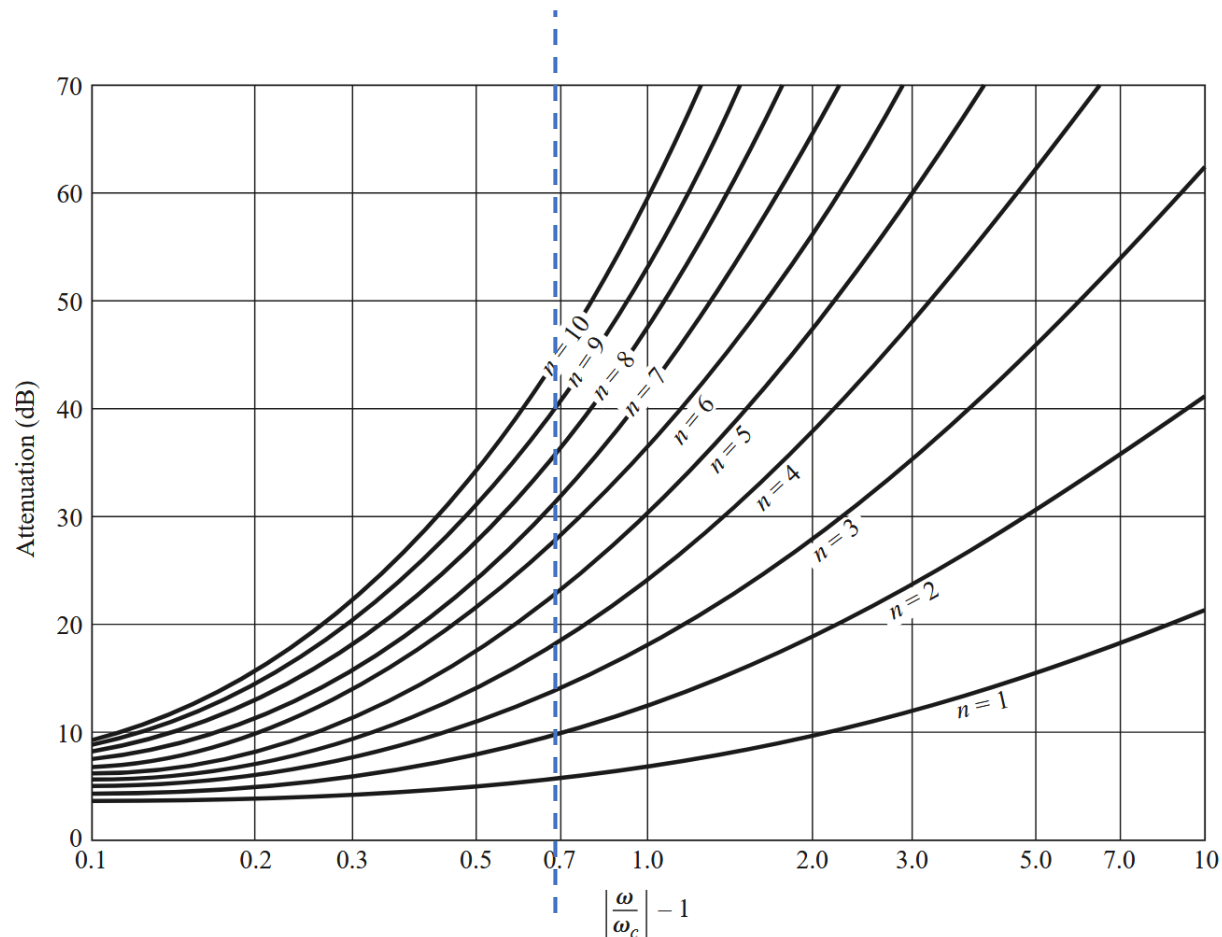


TABLE 8.6 Summary of Prototype Filter Transformations $\left(\Delta = \frac{\omega_2 - \omega_1}{\omega_0}\right)$

Low-pass	High-pass	Bandpass	Bandstop
			
			

例题 设计一个最大平坦响应的LC低通滤波器，通带为 0-2 GHz，在 3.4 GHz 处的衰减至少为 20 dB，特征阻抗为 50 Ω 。

- (1) 给出滤波器的电路结构以及各元件的取值（任取一种电路形式）；
- (2) 使用 ADS 绘制插入损耗（S21）随频率变化的幅值曲线（dB）。



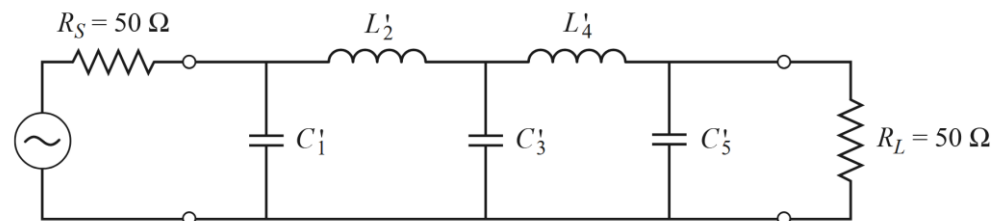
$$\left| \frac{\omega}{\omega_c} \right| - 1 = 0.7$$

$$N = 5$$

TABLE 8.3 Element Values for Maximally Flat Low-Pass Filter Prototypes ($g_0 = 1$, $\omega_c = 1$, $N = 1$ to 10)

N	g_1	g_2	g_3	g_4	g_5	g_6	g_7	g_8	g_9	g_{10}	g_{11}
1	2.0000	1.0000									
2	1.4142	1.4142	1.0000								
3	1.0000	2.0000	1.0000	1.0000							
4	0.7654	1.8478	1.8478	0.7654	1.0000						
5	0.6180	1.6180	2.0000	1.6180	0.6180	1.0000					
6	0.5176	1.4142	1.9318	1.9318	1.4142	0.5176	1.0000				
7	0.4450	1.2470	1.8019	2.0000	1.8019	1.2470	0.4450	1.0000			
8	0.3902	1.1111	1.6629	1.9615	1.9615	1.6629	1.1111	0.3902	1.0000		
9	0.3473	1.0000	1.5321	1.8794	2.0000	1.8794	1.5321	1.0000	0.3473	1.0000	
10	0.3129	0.9080	1.4142	1.7820	1.9754	1.9754	1.7820	1.4142	0.9080	0.3129	1.0000

Source: Reprinted from G. L. Matthaei, L. Young, and E. M. T. Jones, *Microwave Filters, Impedance-Matching Networks, and Coupling Structures*, Artech House, Dedham, Mass., 1980, with permission.



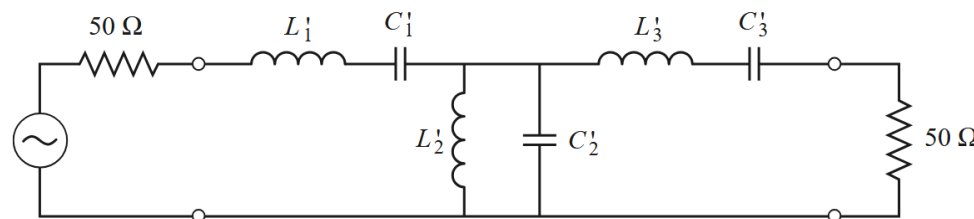
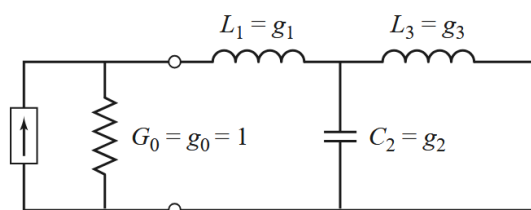
$$L'_k = \frac{g_k R_0}{2\pi f_c}$$

$$C'_k = \frac{g_k}{R_0 2\pi f_c}$$

例题

设计阶数 $N = 3$ 的带通滤波器，中心频率为 1 GHz ，带宽为 10% ，阻抗为 50Ω 。

- 1、要求设计等波纹 0.5 dB 的滤波器，给出电路结构以及各元件的取值（任取一种电路形式）；
- 2、要求设计最大平坦响应滤波器，给出电路结构以及各元件的取值（任取一种电路形式）；
- 3、使用 ADS 画出两种滤波器的 $S_{21} \text{ (dB)}$ 幅值曲线以及群延时曲线（ $0.8\text{-}1.2 \text{ GHz}$ ，步长为 1 MHz ），并比较两种滤波器的特点。



$$\begin{aligned} g_1 &= L_1 = 1 \\ g_2 &= C_2 = 2 \\ g_3 &= L_3 = 1 \end{aligned}$$

Low-pass	High-pass	Bandpass	Bandstop

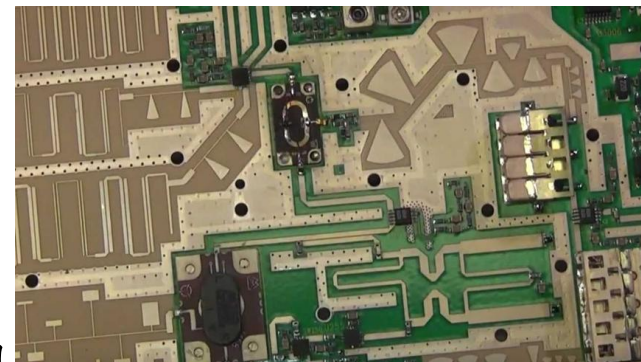
$$\begin{aligned} g_1 &= 1.5963 = L_1, \\ g_2 &= 1.0967 = C_2, \\ g_3 &= 1.5963 = L_3, \\ g_4 &= 1.000 = R_L. \end{aligned}$$



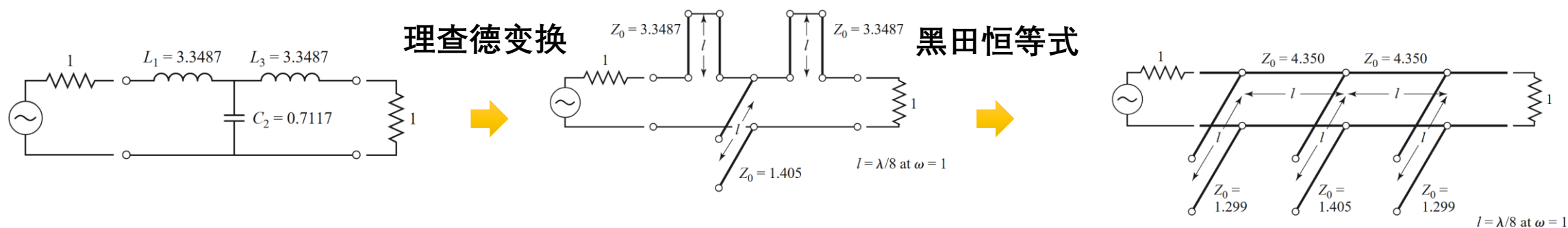
$$\begin{aligned} L'_1 &= \frac{L_1 R_0}{\omega_0 \Delta} = 127.0 \text{ nH}, \\ C'_1 &= \frac{\Delta}{\omega_0 L_1 R_0} = 0.199 \text{ pF}, \\ L'_2 &= \frac{\Delta R_0}{\omega_0 C_2} = 0.726 \text{ nH}, \\ C'_2 &= \frac{C_2}{\omega_0 \Delta R_0} = 34.91 \text{ pF}, \\ L'_3 &= \frac{L_3 R_0}{\omega_0 \Delta} = 127.0 \text{ nH}, \\ C'_3 &= \frac{\Delta}{\omega_0 L_3 R_0} = 0.199 \text{ pF}. \end{aligned}$$

$$\begin{aligned} L'_1 &= \frac{L_1 R_0}{\omega_0 \Delta} = 79.6 \text{ nH} \\ C'_1 &= \frac{L_1 R_0}{\omega_0 \Delta} = 0.318 \text{ pF} \\ L'_2 &= \frac{\Delta R_0}{\omega_0 C_2} = 0.398 \text{ nH} \\ C'_2 &= \frac{C_2}{\omega_0 \Delta R_0} = 63.66 \text{ pF} \\ L'_3 &= \frac{L_3 R_0}{\omega_0 \Delta} = 79.6 \text{ nH} \\ C'_1 &= \frac{L_3 R_0}{\omega_0 \Delta} = 0.318 \text{ pF} \end{aligned}$$

集总转分布式滤波器



- 集总电感和电容只存在有限频带内
- 集总元件之间的连接部分对频带响应有影响
- **理查德变换 (Richards' transformation)** 将集总元件转化为传输线
- **黑田恒等式 (Kuroda's identities)** 使用传输线将滤波器元件组分开



理查德变换

将 ω 域转换成 Ω 域

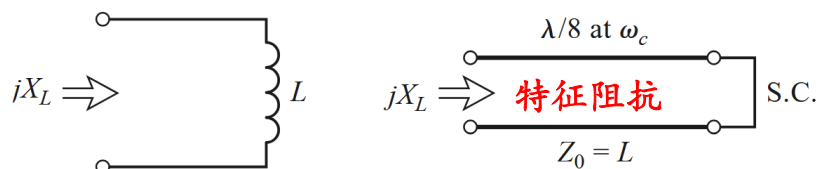
$$\Omega = \frac{\omega}{\omega_c} = \tan \beta l = \tan\left(\frac{\omega l}{v_p}\right) \Rightarrow \begin{cases} \square \text{ 当 } \omega = 0 \text{ 时, } \Omega = 0; \\ \square \text{ 当 } \omega = \omega_c \text{ 时, } \Omega = 1 \text{ (所以 } l = \lambda/8 \text{)} \end{cases}$$



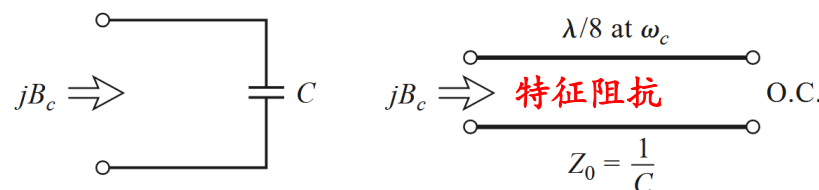
$$jX_L = j\Omega L = j\mathbf{L}\tan \beta l$$

$$jB_C = j\Omega C = j\mathbf{C}\tan \beta l$$

这里的L/C相当于滤波器设计表格中的g值!

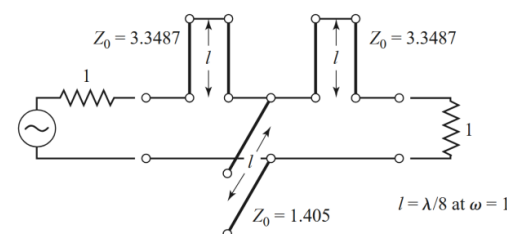
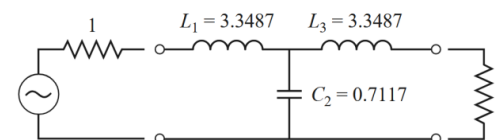


(a)



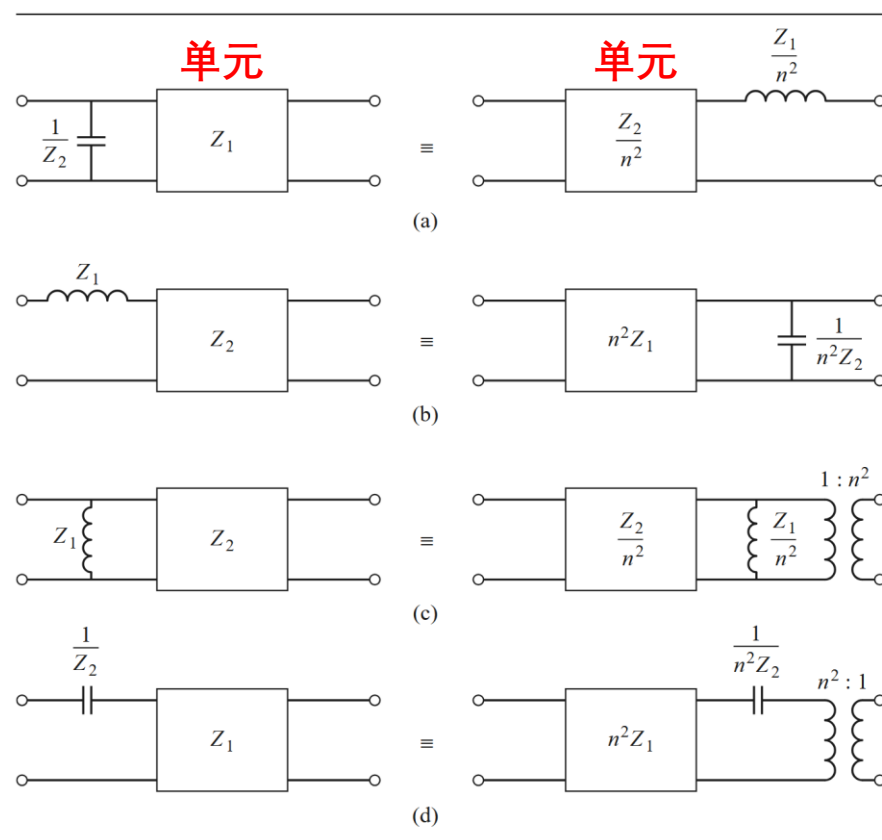
(b)

例如:



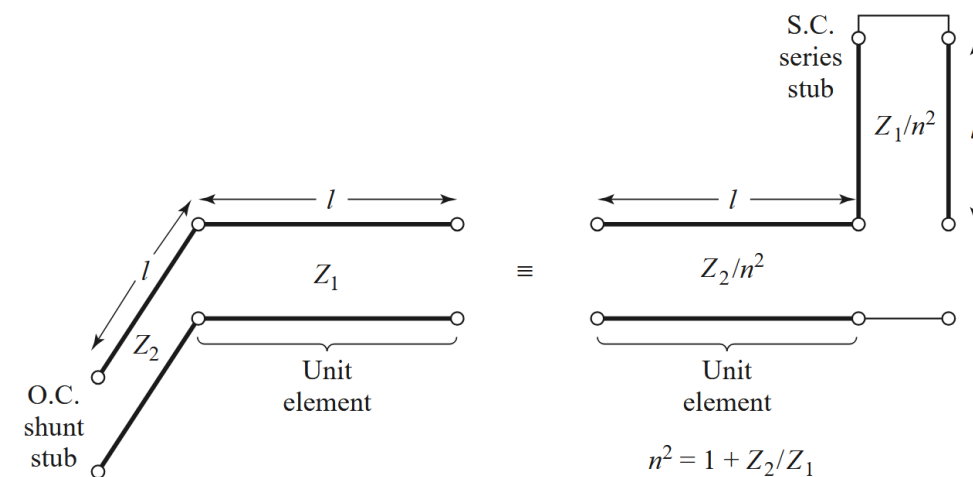
黑田恒等式

TABLE 8.7 The Four Kuroda Identities ($n^2 = 1 + Z_2/Z_1$)



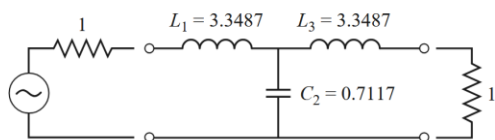
单元：截止频率的1/8波长长度的传输线

以表a恒等式举例：

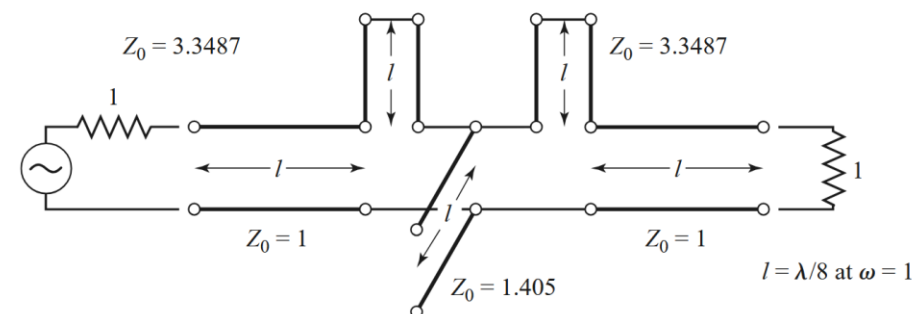
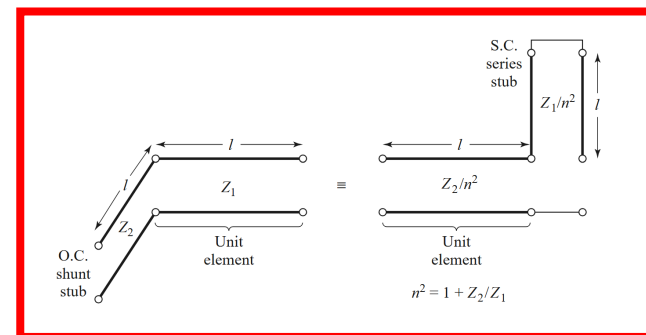
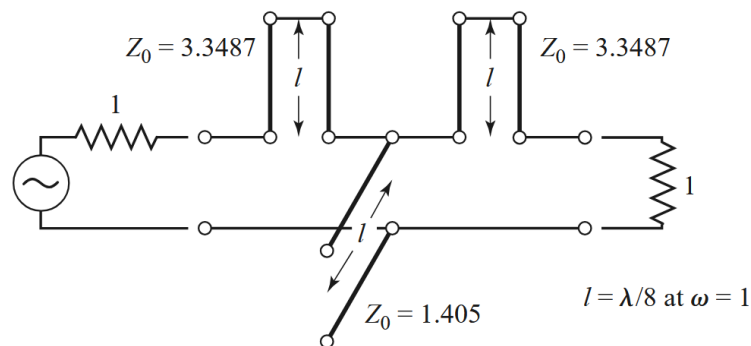


例题

设计基于微带线的低通滤波器，截止频率为4 GHz，阻抗为50 Ω ，要求三阶3 dB等纹波通带响应。

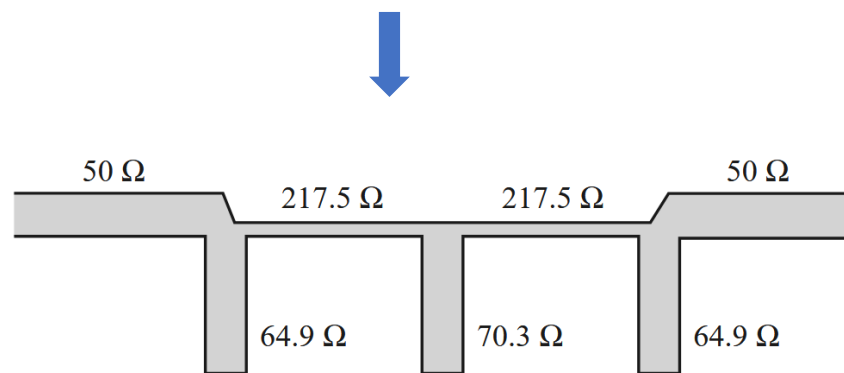
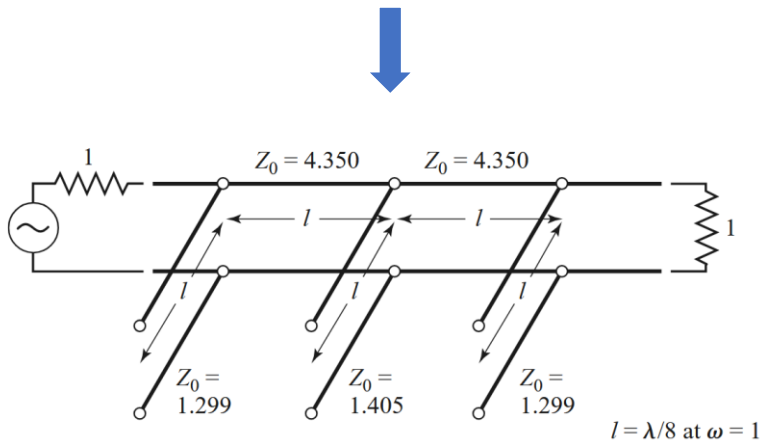
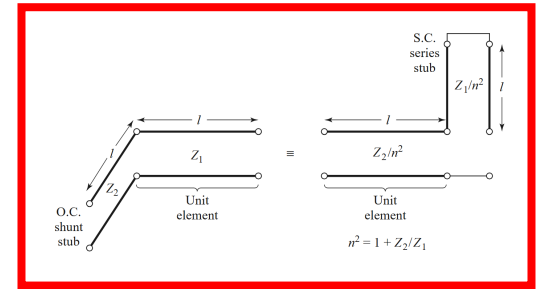
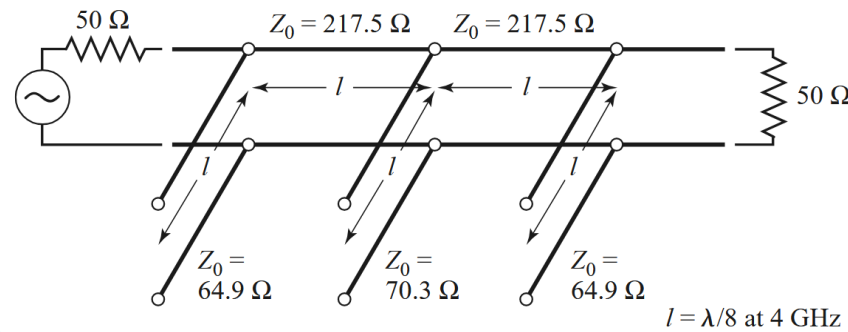
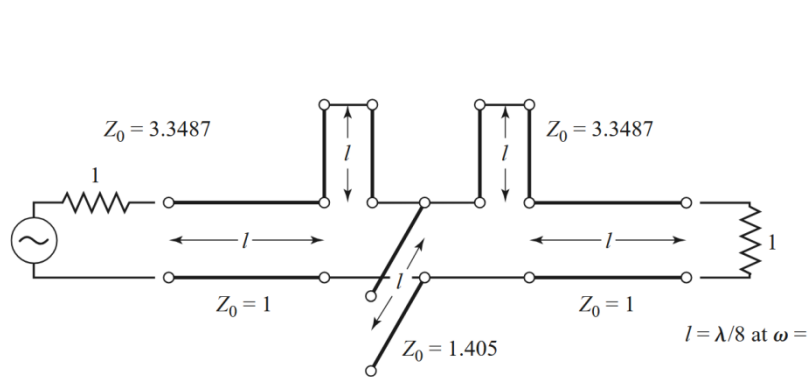


$$\begin{aligned} g_1 &= 3.3487 = L_1, \\ g_2 &= 0.7117 = C_2, \\ g_3 &= 3.3487 = L_3, \\ g_4 &= 1.0000 = R_L, \end{aligned}$$



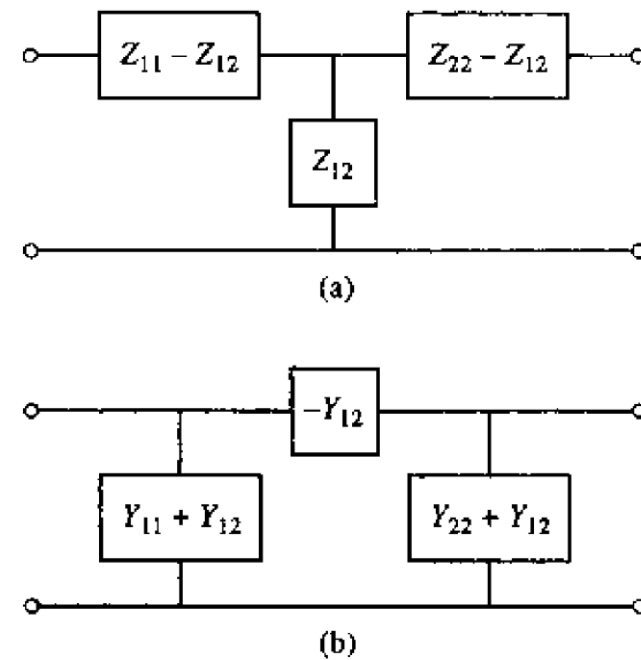
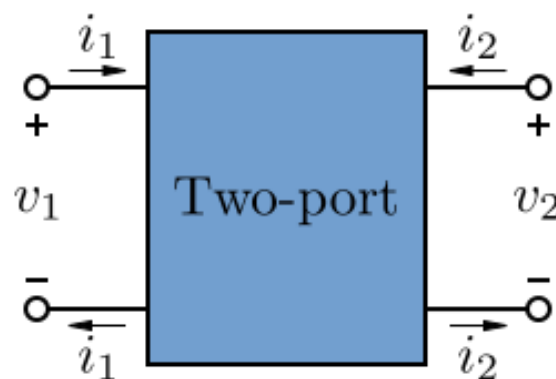
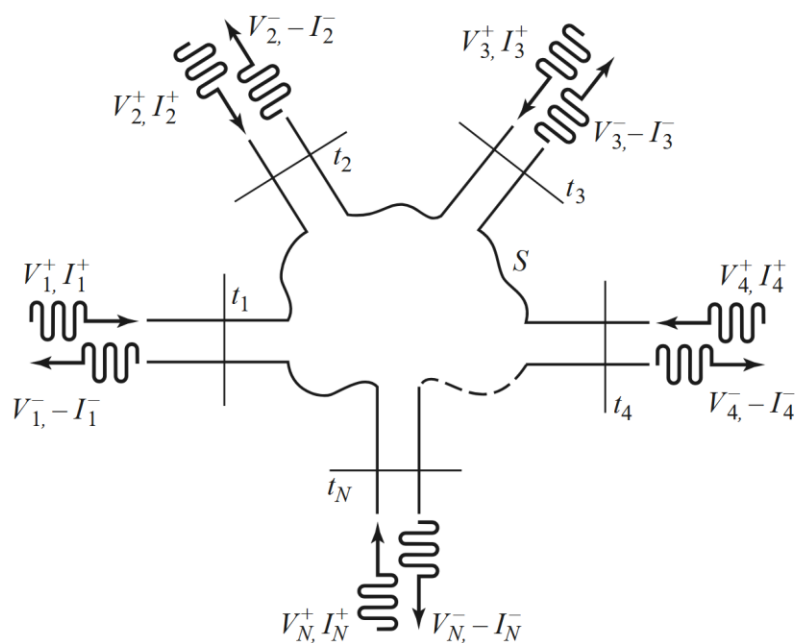
例题

设计基于微带线的低通滤波器，截止频率为4 GHz，阻抗为50 Ω ，要求三阶3 dB等纹波通带响应。



微波网络理论

- S散射矩阵
- Z阻抗矩阵
- ABCD 传输矩阵



微波网络理论

- S散射矩阵
- Z阻抗矩阵
- ABCD 传输矩阵

TABLE 4.1 ABCD Parameters of Some Useful Two-Port Circuits

Circuit	ABCD Parameters	
	$A = 1$ $C = 0$	$B = Z$ $D = 1$
	$A = 1$ $C = Y$	$B = 0$ $D = 1$
	$A = \cos \beta l$ $C = j Y_0 \sin \beta l$	$B = j Z_0 \sin \beta l$ $D = \cos \beta l$
	$A = N$ $C = 0$	$B = 0$ $D = \frac{1}{N}$
	$A = 1 + \frac{Y_2}{Y_3}$ $C = Y_1 + Y_2 + \frac{Y_1 Y_2}{Y_3}$	$B = \frac{1}{Y_3}$ $D = 1 + \frac{Y_1}{Y_3}$
	$A = 1 + \frac{Z_1}{Z_3}$ $C = \frac{1}{Z_3}$	$B = Z_1 + Z_2 + \frac{Z_1 Z_2}{Z_3}$ $D = 1 + \frac{Z_2}{Z_3}$

TABLE 4.2 Conversions Between Two-Port Network Parameters

	S	Z	Y	$ABCD$
S_{11}	S_{11}	$\frac{(Z_{11} - Z_0)(Z_{22} + Z_0) - Z_{12}Z_{21}}{\Delta Z}$	$\frac{(Y_0 - Y_{11})(Y_0 + Y_{22}) + Y_{12}Y_{21}}{\Delta Y}$	$\frac{A + B/Z_0 - CZ_0 - D}{A + B/Z_0 + CZ_0 + D}$
S_{12}	S_{12}	$\frac{2Z_{12}Z_0}{\Delta Z}$	$\frac{-2Y_{12}Y_0}{\Delta Y}$	$\frac{2(AD - BC)}{A + B/Z_0 + CZ_0 + D}$
S_{21}	S_{21}	$\frac{2Z_{21}Z_0}{\Delta Z}$	$\frac{-2Y_{21}Y_0}{\Delta Y}$	$\frac{2}{A + B/Z_0 + CZ_0 + D}$
S_{22}	S_{22}	$\frac{(Z_{11} + Z_0)(Z_{22} - Z_0) - Z_{12}Z_{21}}{\Delta Z}$	$\frac{(Y_0 + Y_{11})(Y_0 - Y_{22}) + Y_{12}Y_{21}}{\Delta Y}$	$\frac{-A + B/Z_0 - CZ_0 + D}{A + B/Z_0 + CZ_0 + D}$
Z_{11}	$Z_0 \frac{(1 + S_{11})(1 - S_{22}) + S_{12}S_{21}}{(1 - S_{11})(1 - S_{22}) - S_{12}S_{21}}$	Z_{11}	$\frac{Y_{22}}{ Y }$	$\frac{A}{C}$
Z_{12}	$Z_0 \frac{2S_{12}}{(1 - S_{11})(1 - S_{22}) - S_{12}S_{21}}$	Z_{12}	$\frac{-Y_{12}}{ Y }$	$\frac{AD - BC}{C}$
Z_{21}	$Z_0 \frac{2S_{21}}{(1 - S_{11})(1 - S_{22}) - S_{12}S_{21}}$	Z_{21}	$\frac{-Y_{21}}{ Y }$	$\frac{1}{C}$
Z_{22}	$Z_0 \frac{(1 - S_{11})(1 + S_{22}) + S_{12}S_{21}}{(1 - S_{11})(1 - S_{22}) - S_{12}S_{21}}$	Z_{22}	$\frac{Y_{11}}{ Y }$	$\frac{D}{C}$
Y_{11}	$Y_0 \frac{(1 - S_{11})(1 + S_{22}) + S_{12}S_{21}}{(1 + S_{11})(1 + S_{22}) - S_{12}S_{21}}$	$\frac{Z_{22}}{ Z }$	Y_{11}	$\frac{D}{B}$
Y_{12}	$Y_0 \frac{-2S_{12}}{(1 + S_{11})(1 + S_{22}) - S_{12}S_{21}}$	$\frac{-Z_{12}}{ Z }$	Y_{12}	$\frac{BC - AD}{B}$
Y_{21}	$Y_0 \frac{-2S_{21}}{(1 + S_{11})(1 + S_{22}) - S_{12}S_{21}}$	$\frac{-Z_{21}}{ Z }$	Y_{21}	$\frac{-1}{B}$
Y_{22}	$Y_0 \frac{(1 + S_{11})(1 - S_{22}) + S_{12}S_{21}}{(1 + S_{11})(1 + S_{22}) - S_{12}S_{21}}$	$\frac{Z_{11}}{ Z }$	Y_{22}	$\frac{A}{B}$
A	$\frac{(1 + S_{11})(1 - S_{22}) + S_{12}S_{21}}{2S_{21}}$	$\frac{Z_{11}}{Z_{21}}$	$\frac{-Y_{22}}{Y_{21}}$	A
B	$Z_0 \frac{(1 + S_{11})(1 + S_{22}) - S_{12}S_{21}}{2S_{21}}$	$\frac{ Z }{Z_{21}}$	$\frac{-1}{Y_{21}}$	B
C	$\frac{1}{Z_0} \frac{(1 - S_{11})(1 - S_{22}) - S_{12}S_{21}}{2S_{21}}$	$\frac{1}{Z_{21}}$	$\frac{- Y }{Y_{21}}$	C
D	$\frac{(1 - S_{11})(1 + S_{22}) + S_{12}S_{21}}{2S_{21}}$	$\frac{Z_{22}}{Z_{21}}$	$\frac{-Y_{11}}{Y_{21}}$	D

$$|Z| = Z_{11}Z_{22} - Z_{12}Z_{21}; \quad |Y| = Y_{11}Y_{22} - Y_{12}Y_{21}; \quad \Delta Y = (Y_{11} + Y_0)(Y_{22} + Y_0) - Y_{12}Y_{21}; \quad \Delta Z = (Z_{11} + Z_0)(Z_{22} + Z_0) - Z_{12}Z_{21}; \quad Y_0 = 1/Z_0.$$

阶梯阻抗型低通滤波器

- Stepped-impedance, or hi-Z , low-Z filters
- 滤波器带外抑制效果一般
- 体积小、易设计

$$Z_{11} - Z_{12} = -jZ_0 \left(\frac{\cos \beta \ell - 1}{\sin \beta \ell} \right) = jZ_0 \tan \left(\frac{\beta \ell}{2} \right),$$

$$Z_{11} = Z_{22} = \frac{A}{C} = -jZ_0 \cot \beta \ell,$$

$$Z_{12} = Z_{21} = \frac{1}{C} = -jZ_0 \csc \beta \ell.$$

TABLE 4.1 ABCD Parameters of Some Useful Two-Port Circuits

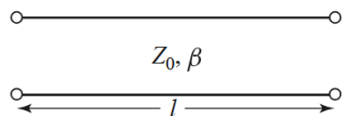
Circuit	ABCD Parameters	
	$A = 1$ $C = 0$	$B = Z$ $D = 1$
	$A = 1$ $C = Y$	$B = 0$ $D = 1$
	$A = \cos \beta \ell$ $C = jY_0 \sin \beta \ell$	$B = jZ_0 \sin \beta \ell$ $D = \cos \beta \ell$
	$A = N$ $C = 0$	$B = 0$ $D = \frac{1}{N}$
	$A = 1 + \frac{Y_2}{Y_3}$ $C = Y_1 + Y_2 + \frac{Y_1 Y_2}{Y_3}$	$B = \frac{1}{Y_3}$ $D = 1 + \frac{Y_1}{Y_3}$
	$A = 1 + \frac{Z_1}{Z_3}$ $C = \frac{1}{Z_3}$	$B = Z_1 + Z_2 + \frac{Z_1 Z_2}{Z_3}$ $D = 1 + \frac{Z_2}{Z_3}$

TABLE 4.2 Conversions Between Two-Port Network Parameters

	S	Z	Y	ABCD
s_{11}	s_{11}	$\frac{(Z_{11} - Z_0)(Z_{22} + Z_0) - Z_{12}Z_{21}}{\Delta Z}$	$\frac{(Y_0 - Y_{11})(Y_0 + Y_{22}) + Y_{12}Y_{21}}{\Delta Y}$	$\frac{A + B/Z_0 - CZ_0 - D}{A + B/Z_0 + CZ_0 + D}$
s_{12}	s_{12}	$\frac{2Z_{12}Z_0}{\Delta Z}$	$\frac{-2Y_{12}Y_0}{\Delta Y}$	$\frac{2(AD - BC)}{A + B/Z_0 + CZ_0 + D}$
s_{21}	s_{21}	$\frac{2Z_{21}Z_0}{\Delta Z}$	$\frac{-2Y_{21}Y_0}{\Delta Y}$	$\frac{2}{A + B/Z_0 + CZ_0 + D}$
s_{22}	s_{22}	$\frac{(Z_{11} + Z_0)(Z_{22} - Z_0) - Z_{12}Z_{21}}{\Delta Z}$	$\frac{(Y_0 + Y_{11})(Y_0 - Y_{22}) + Y_{12}Y_{21}}{\Delta Y}$	$\frac{-A + B/Z_0 - CZ_0 + D}{A + B/Z_0 + CZ_0 + D}$
Z_{11}	$Z_0 \frac{(1 + s_{11})(1 - s_{22}) + s_{12}s_{21}}{(1 - s_{11})(1 - s_{22}) - s_{12}s_{21}}$	Z_{11}	$\frac{Y_{22}}{ Y }$	$\frac{A}{C}$
Z_{12}	$Z_0 \frac{2s_{12}}{(1 - s_{11})(1 - s_{22}) - s_{12}s_{21}}$	Z_{12}	$\frac{-Y_{12}}{ Y }$	$\frac{AD - BC}{C}$
Z_{21}	$Z_0 \frac{2s_{21}}{(1 - s_{11})(1 - s_{22}) - s_{12}s_{21}}$	Z_{21}	$\frac{-Y_{21}}{ Y }$	$\frac{1}{C}$
Z_{22}	$Z_0 \frac{(1 - s_{11})(1 + s_{22}) + s_{12}s_{21}}{(1 - s_{11})(1 - s_{22}) - s_{12}s_{21}}$	Z_{22}	$\frac{Y_{11}}{ Y }$	$\frac{D}{C}$
Y_{11}	$Y_0 \frac{(1 - s_{11})(1 + s_{22}) + s_{12}s_{21}}{(1 + s_{11})(1 + s_{22}) - s_{12}s_{21}}$	$\frac{Z_{22}}{ Z }$	Y_{11}	$\frac{D}{B}$
Y_{12}	$Y_0 \frac{-2s_{12}}{(1 + s_{11})(1 + s_{22}) - s_{12}s_{21}}$	$\frac{-Z_{12}}{ Z }$	Y_{12}	$\frac{BC - AD}{B}$
Y_{21}	$Y_0 \frac{-2s_{21}}{(1 + s_{11})(1 + s_{22}) - s_{12}s_{21}}$	$\frac{-Z_{21}}{ Z }$	Y_{21}	$\frac{-1}{B}$
Y_{22}	$Y_0 \frac{(1 + s_{11})(1 - s_{22}) + s_{12}s_{21}}{(1 + s_{11})(1 + s_{22}) - s_{12}s_{21}}$	$\frac{Z_{11}}{ Z }$	Y_{22}	$\frac{A}{B}$
A	$\frac{(1 + s_{11})(1 - s_{22}) + s_{12}s_{21}}{2s_{21}}$	$\frac{Z_{11}}{Z_{21}}$	$\frac{-Y_{22}}{Y_{21}}$	A
B	$Z_0 \frac{(1 + s_{11})(1 + s_{22}) - s_{12}s_{21}}{2s_{21}}$	$\frac{ Z }{Z_{21}}$	$\frac{-1}{Y_{21}}$	B
C	$\frac{1}{Z_0} \frac{(1 - s_{11})(1 - s_{22}) - s_{12}s_{21}}{2s_{21}}$	$\frac{1}{Z_{21}}$	$\frac{- Y }{Y_{21}}$	C
D	$\frac{(1 - s_{11})(1 + s_{22}) + s_{12}s_{21}}{2s_{21}}$	$\frac{Z_{22}}{Z_{21}}$	$\frac{-Y_{11}}{Y_{21}}$	D

$$|Z| = Z_{11}Z_{22} - Z_{12}Z_{21}; \quad |Y| = Y_{11}Y_{22} - Y_{12}Y_{21}; \quad \Delta Y = (Y_{11} + Y_0)(Y_{22} + Y_0) - Y_{12}Y_{21}; \quad \Delta Z = (Z_{11} + Z_0)(Z_{22} + Z_0) - Z_{12}Z_{21}; \quad Y_0 = 1/Z_0.$$

阶梯阻抗型低通滤波器

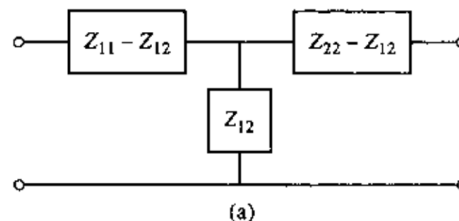


$$A = \cos \beta l$$

$$C = j Y_0 \sin \beta l$$

$$B = j Z_0 \sin \beta l$$

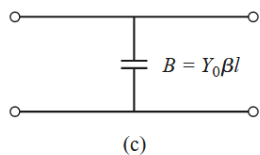
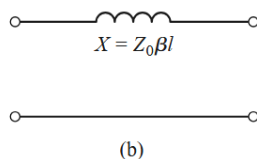
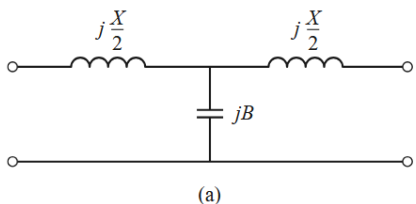
$$D = \cos \beta l$$



$$Z_{11} - Z_{12} = -j Z_0 \left(\frac{\cos \beta l - 1}{\sin \beta l} \right) = j Z_0 \tan \left(\frac{\beta l}{2} \right),$$

$$Z_{11} = Z_{22} = \frac{A}{C} = -j Z_0 \cot \beta l,$$

$$Z_{12} = Z_{21} = \frac{1}{C} = -j Z_0 \csc \beta l.$$



□ 当 $\beta l < \pi/4$, 特征阻抗较大时, $Z_0 = Z_h$

$$\begin{aligned} X &\approx Z_h \beta l \\ B &\approx 0 \end{aligned} \Rightarrow L'_k = \frac{R_0 L_k (g_i)}{\omega_c} \Rightarrow \beta l = \frac{L R_0}{Z_h} \quad (\text{inductor})$$

□ 当 $\beta l < \pi/4$, 特征阻抗较小时, $Z_0 = Z_l$

$$\frac{X}{2} = Z_0 \tan \left(\frac{\beta l}{2} \right),$$

$$B = \frac{1}{Z_0} \sin \beta l.$$

$$\begin{aligned} X &\approx 0 \\ B &\approx \frac{1}{Z_l} \beta l \end{aligned} \Rightarrow C'_k = \frac{C_k (g_i)}{R_0 \omega_c} \Rightarrow \beta l = \frac{C Z_l}{R_0} \quad (\text{capacitor}),$$

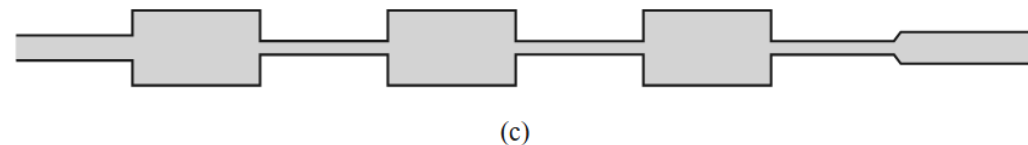
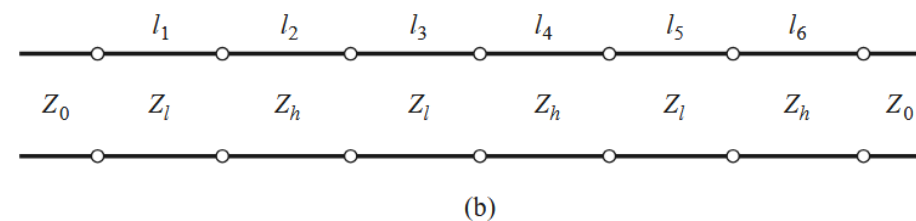
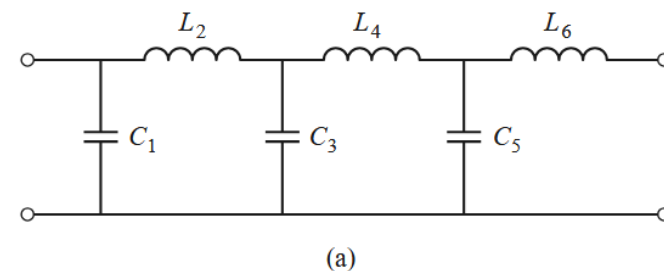
例题

设计一个阶梯阻抗低通滤波器，其响应为最大平坦型，截止频率为 2.5 GHz。要求在 4 GHz 时的插入损耗大于 20 dB。滤波器的特性阻抗为 $50\ \Omega$ ；实际可实现的最高传输线阻抗为 $120\ \Omega$ ，最低为 $20\ \Omega$ 。考虑当该滤波器采用微带线基板实现时的损耗影响，其中基板参数为：厚度 $d = 0.158\text{ cm}$ ，相对介电常数 $\epsilon_r = 4.2$ ，损耗角正切 $\tan\delta = 0.02$ ，以及铜导体厚度为 0.5 mil。(1 mil=0.0254 mm)

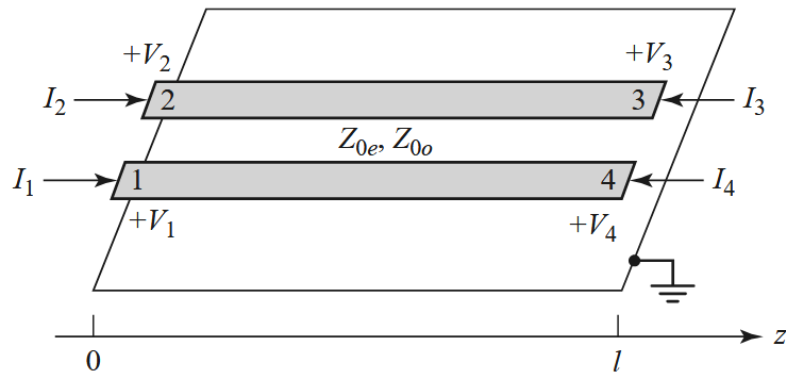
$$N = 6$$

$$\begin{aligned} g_1 &= 0.517 = C_1, \\ g_2 &= 1.414 = L_2, \\ g_3 &= 1.932 = C_3, \\ g_4 &= 1.932 = L_4, \\ g_5 &= 1.414 = C_5, \\ g_6 &= 0.517 = L_6. \end{aligned}$$

Section	$Z_i = Z_\ell \text{ or } Z_h(\Omega)$	$\beta\ell_i \text{ (deg)}$	$W_i \text{ (mm)}$	$\ell_i \text{ (mm)}$
1	20	11.8	11.3	2.05
2	120	33.8	0.428	6.63
3	20	44.3	11.3	7.69
4	120	46.1	0.428	9.04
5	20	32.4	11.3	5.63
6	120	12.3	0.428	2.41

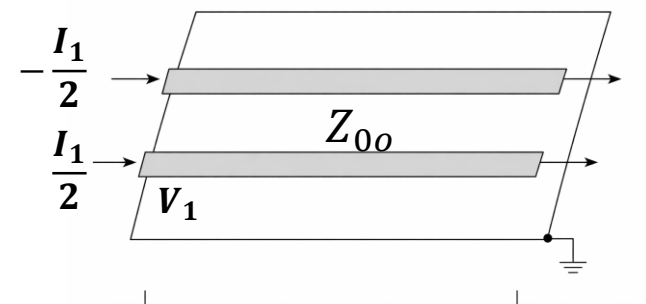


耦合线型滤波器



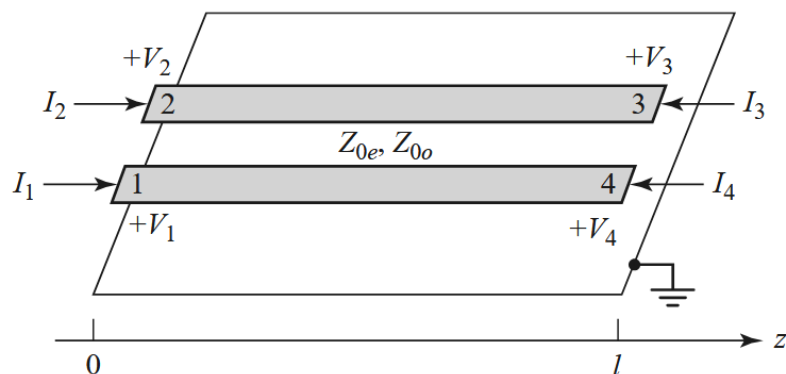
$$\begin{pmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{pmatrix} = \begin{pmatrix} Z_{11} & Z_{12} & Z_{13} & Z_{14} \\ Z_{21} & Z_{22} & Z_{23} & Z_{24} \\ Z_{31} & Z_{32} & Z_{33} & Z_{34} \\ Z_{41} & Z_{42} & Z_{43} & Z_{44} \end{pmatrix} \begin{pmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{pmatrix}$$

耦合结构需要解耦，有效方法为奇偶模分解法/even- and odd-mode decomposition method



$$Z_{11} = Z_{22} = Z_{33} = Z_{44} = \frac{-j}{2} (Z_{0e} + Z_{0o}) \cot \theta \quad \theta = \beta l$$

耦合线型滤波器



$$\begin{pmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{pmatrix} = \begin{pmatrix} Z_{11} & Z_{12} & Z_{13} & Z_{14} \\ Z_{21} & Z_{22} & Z_{23} & Z_{24} \\ Z_{31} & Z_{32} & Z_{33} & Z_{34} \\ Z_{41} & Z_{42} & Z_{43} & Z_{44} \end{pmatrix} \begin{pmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{pmatrix}$$

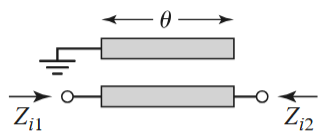
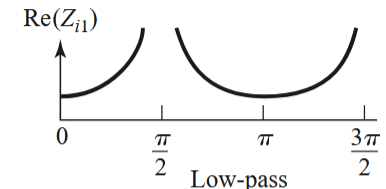
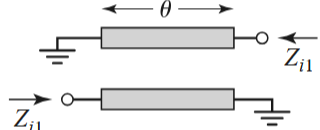
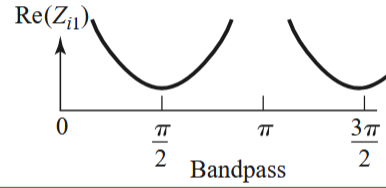
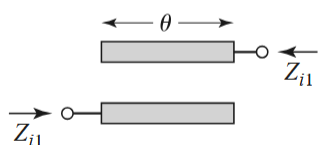
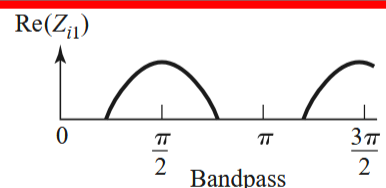
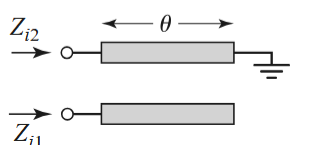
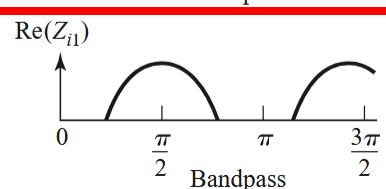
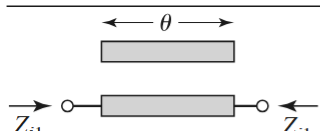
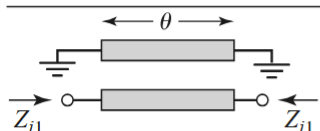
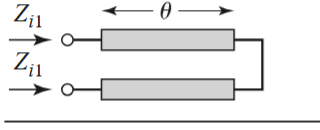
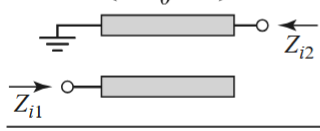
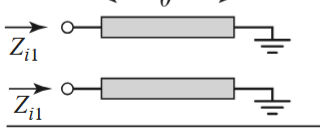
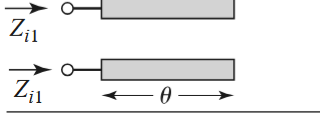
$$Z_{11} = Z_{22} = Z_{33} = Z_{44} = \frac{-j}{2} (Z_{0e} + Z_{0o}) \cot \theta$$

$$Z_{12} = Z_{21} = Z_{34} = Z_{43} = \frac{-j}{2} (Z_{0e} - Z_{0o}) \cot \theta$$

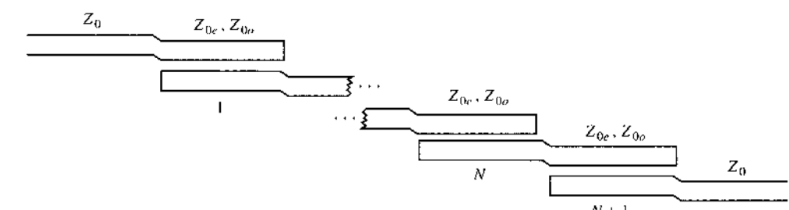
$$Z_{13} = Z_{31} = Z_{24} = Z_{42} = \frac{-j}{2} (Z_{0e} - Z_{0o}) \csc \theta$$

$$Z_{14} = Z_{41} = Z_{23} = Z_{32} = \frac{-j}{2} (Z_{0e} + Z_{0o}) \csc \theta$$

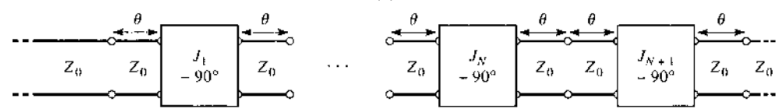
不同端口设置下的滤波器类型

Circuit	Image Impedance	Response
	$Z_{i1} = \frac{2Z_{0e}Z_{0o} \cos \theta}{\sqrt{(Z_{0e} + Z_{0o})^2 \cos^2 \theta - (Z_{0e} - Z_{0o})^2}}$ $Z_{i2} = \frac{Z_{0e}Z_{0o}}{Z_{i1}}$	 Low-pass
	$Z_{i1} = \frac{2Z_{0e}Z_{0o} \sin \theta}{\sqrt{(Z_{0e} - Z_{0o})^2 - (Z_{0e} + Z_{0o})^2 \cos^2 \theta}}$	 Bandpass
	$Z_{i1} = \frac{\sqrt{(Z_{0e} - Z_{0o})^2 - (Z_{0e} + Z_{0o})^2 \cos^2 \theta}}{2 \sin \theta}$	 Bandpass
	$Z_{i1} = \frac{\sqrt{Z_{0e}Z_{0o}} \sqrt{(Z_{0e} - Z_{0o})^2 - (Z_{0e} + Z_{0o})^2 \cos^2 \theta}}{(Z_{0e} + Z_{0o}) \sin \theta}$ $Z_{i2} = \frac{Z_{0e}Z_{0o}}{Z_{i1}}$	 Bandpass
	$Z_{i1} = \frac{Z_{0e} + Z_{0o}}{2}$	All pass
	$Z_{i1} = \frac{2Z_{0e}Z_{0o}}{Z_{0e} + Z_{0o}}$	All pass
	$Z_{i1} = \sqrt{Z_{0e}Z_{0o}}$	All pass
	$Z_{i1} = -j \frac{2Z_{0e}Z_{0o}}{Z_{0e} + Z_{0o}} \cot \theta$ $Z_{i2} = \frac{Z_{0e}Z_{0o}}{Z_{i1}}$	All stop
	$Z_{i1} = j \sqrt{Z_{0e}Z_{0o}} \tan \theta$	All stop
	$Z_{i1} = -j \sqrt{Z_{0e}Z_{0o}} \cot \theta$	All stop

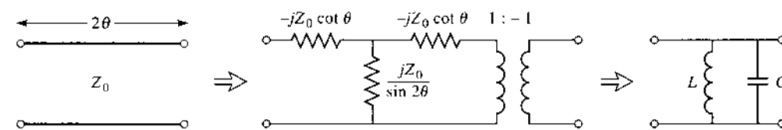
耦合型传输线的带通滤波器



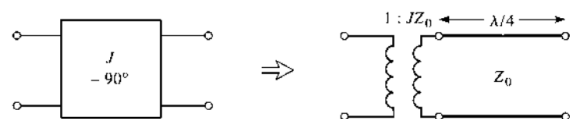
(a)



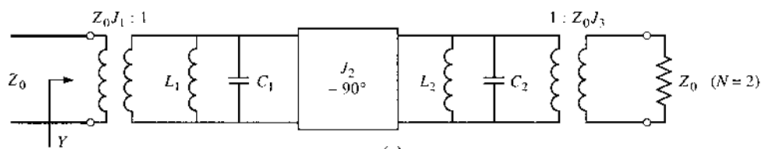
(b)



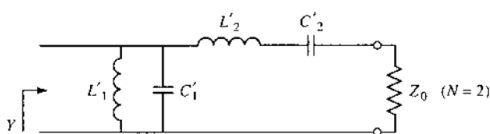
(c)



(d)

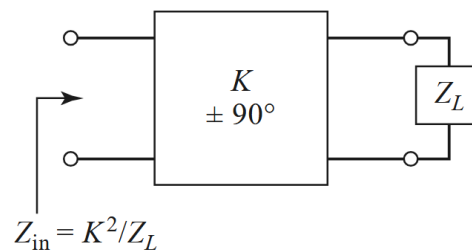


(e)

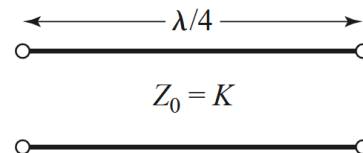


(f)

Impedance inverters

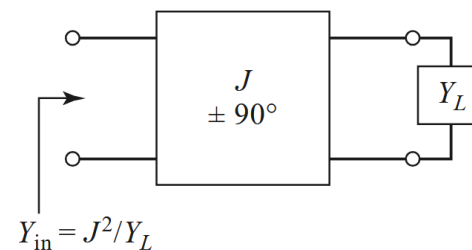


(a)

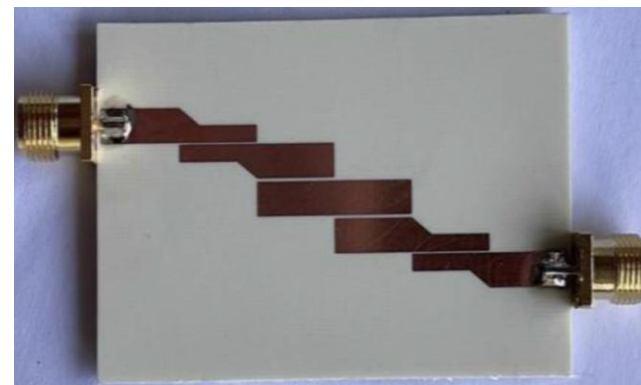
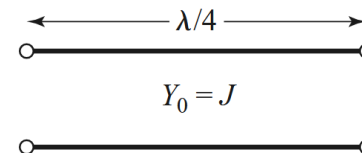


(b)

Admittance inverters



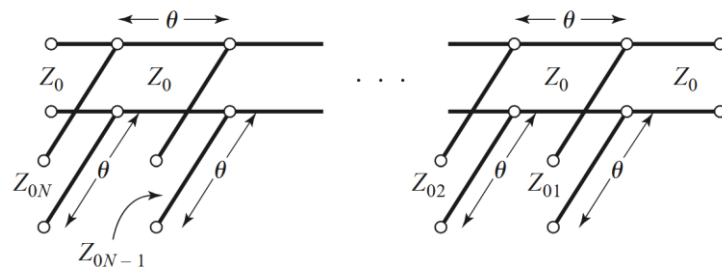
(a)



基于 $\lambda/4$ 谐振器的带阻/带通滤波器

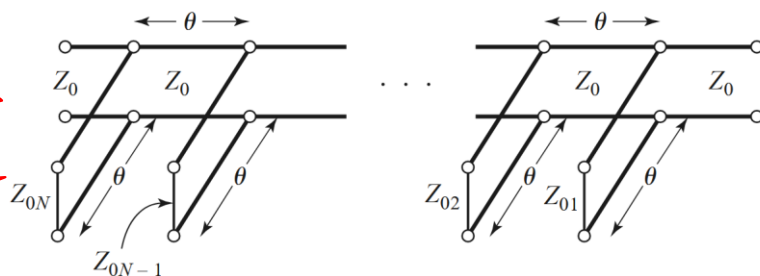
$$\theta = \pi/2$$

带阻

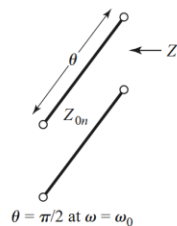


$$Z_{0n} = \frac{4Z_0}{\pi g_n \Delta}$$

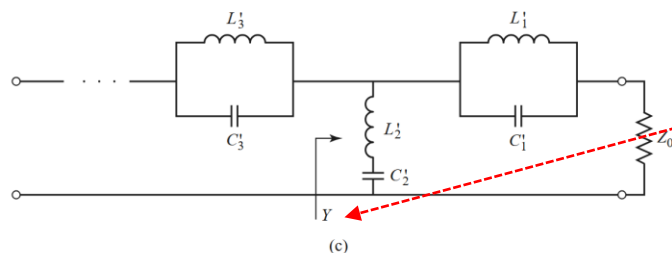
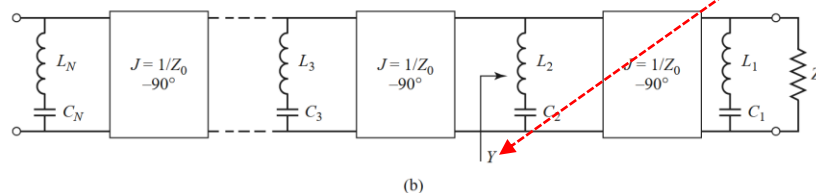
带通



$$Z_{0n} = \frac{\pi Z_0 \Delta}{4g_n}$$



$$Z_{0n} = \frac{4\omega_0 L_n}{\pi}$$



$$Y = \frac{1}{j\omega L_2 + (1/j\omega C_2)} + \frac{1}{Z_0^2} \left(\frac{1}{j\omega L_1 + 1/j\omega C_1} + \frac{1}{Z_0} \right)^{-1}$$

$$= \frac{1}{j\sqrt{L_2/C_2}[(\omega/\omega_0) - (\omega_0/\omega)]} + \frac{1}{Z_0} \left\{ \frac{1}{j\sqrt{L_1/C_1}[(\omega/\omega_0) - (\omega_0/\omega)]} + \frac{1}{Z_0} \right\}^{-1}$$

$$Y = \frac{1}{j\omega L'_2 + 1/j\omega C'_2} + \left(\frac{1}{j\omega C'_1 + 1/j\omega L'_1} + Z_0 \right)^{-1}$$

$$= \frac{1}{j\sqrt{L'_2/C'_2}[(\omega/\omega_0) - (\omega_0/\omega)]} + \left\{ \frac{1}{j\sqrt{C'_1/L'_1}[(\omega/\omega_0) - (\omega_0/\omega)]} + Z_0 \right\}^{-1}$$

$$L_1 = \frac{Z_0^2}{\omega_0^2 L'_1}, \quad Z_{0n} = \frac{4Z_0}{\pi g_n \Delta}$$

$$L_2 = L'_2.$$

例题

使用三段四分之一波长开路短截线设计一个带阻滤波器。中心频率为2.0 GHz，带宽为15%，阻抗为50 Ω 。采用等波纹响应，波纹电平为0.5 dB。

n	g_n	$Z_{0n}(\Omega)$
1	1.5963	265.9
2	1.0967	387.0
3	1.5963	265.9